

A
MAJOR PROJECT REPORT
ON
RAG Enhanced Global Assistant for Agriculture Pest and Disease
Management

Submitted in partial fulfillment of the requirements
for the award of Degree of

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the Major Project Work entitled “**RAG Enhanced Global Assistant for Agriculture Pest and Disease Management**” submitted by **G.V.R. Vineela (2453-22-733-143), G. Sai Pranitha (2453-22-733-148), Rishit Senapati (2453-22-733-176)** is a Bonafide work carried out in partial fulfillment for the award of the degree of **Bachelor of Engineering in Computer Science and Engineering** from Neil Gogte Institute of Technology, affiliated to Osmania University, Hyderabad during the academic year **2022-2026**.

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DECLARATION

We declare that the Major Project entitled, “**RAG Enhanced Global Assistant for Agriculture Pest and Disease Management**” is done by us under the guidance of Mrs. K. J. Archana, Assistant Professor, Department of Computer Science and Engineering, Neil Gogte Institute of Technology, affiliated to Osmania University.

Further the results embodied in this project have not been submitted to any other University or Institute for the award of any degree.

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LIST OF ABBREVIATIONS

S.No.	Acronym	Full Form
1.	AI	Artificial Intelligence
2.	ML	Machine Learning
3.	DL	Deep Learning
4.	RAG	Retrieval Augmented Generation
5.	LMM	Large Multimodal Model
6.	LLM	Large Language Model
7.	IoT	Internet of Things
8.	CNN	Convolutional Neural Network
9.	VLM	Vision Language Model
10.	FAISS	Facebook AI Similarity Search
11.	NLP	Natural Language Processing
12.	NLLB	No Language Left Behind
13.	CPU	Central Processing Unit
14.	GPU	Graphics Processing Unit
15.	RAM	Random Access Memory
16.	SSD	Solid State Drive
17.	SVM	Support Vector Machine
18.	KNN	K-Nearest Neighbors
19.	DB	Database
20.	API	Application Programming Interface

ABSTRACT

Agricultural productivity is often affected by crop pests and diseases, which require timely and accurate identification to reduce yield losses and ensure food security. In many areas, farmers depend on manual inspection and traditional practices, which may not always be effective for identifying complex or new plant diseases. With the rapid growth of artificial intelligence, large multimodal models (LMMs), such as Agri-LLaVA, have shown strong potential in automating pest and disease detection by combining visual (image) and textual information. These models can analyze crop images and provide useful descriptions and recommendations, reducing the need for constant expert involvement. However, despite these benefits, Agri-LLaVA has some limitations. It may generate inaccurate or misleading outputs due to hallucinations, where the model produces information that is not factually correct. Additionally, limited and imbalanced training datasets reduce its ability to generalize well across different crops and environmental conditions, affecting its reliability in real-world situations.

To overcome these challenges, this study proposes an improved agricultural assistant that integrates Retrieval-Augmented Generation (RAG). This approach allows the model to retrieve relevant information from reliable agricultural knowledge sources before generating responses, ensuring that outputs are accurate and trustworthy. As a result, misinformation is reduced, and the system becomes more dependable for practical use. The study also focuses on improving the dataset by collecting and incorporating diverse real-world images from multiple sources, covering various crops, diseases, pests, and environmental conditions. The proposed system not only detects crop diseases more accurately but also provides actionable suggestions for treatment and prevention. Experimental results show that the enhanced model performs better than the baseline Agri-LLaVA in terms of accuracy, reliability, and consistency. This research supports the development of scalable and intelligent AI-based agricultural solutions that can assist farmers, experts, and policymakers. Overall, it contributes to sustainable agriculture by reducing crop losses and improving productivity.

KeyWords: Pest and disease detection, Multimodal model, Retrieval-Augmented Generation (RAG), Dataset expansion, Crop disease diagnosis, Hallucination reduction, Image-text analysis, Precision farming, Large Language Model (LLM), Plant health monitoring, Computer vision in agriculture.

INTRODUCTION

1.1 Outlines of the topic

Agriculture is the backbone of many economies, especially in developing countries like India. It plays a crucial role in providing food, employment, and raw materials for industries. A large portion of the population depends directly or indirectly on agriculture for their livelihood. However, agricultural productivity is facing several challenges due to changing environmental conditions, climate variability, and increasing population demands.

One of the major challenges in agriculture is the occurrence of crop pests and diseases. These biological factors significantly affect crop growth, yield, and quality. Pests such as insects and microorganisms like bacteria, fungi, and viruses attack crops at different stages of growth. If not detected early, these diseases can spread rapidly and cause severe damage, leading to economic losses for farmers and affecting food supply.

According to agricultural studies, nearly 20–40% of global crop production is lost every year due to pests and diseases. This highlights the need for effective monitoring and early detection systems. Traditional methods of disease identification rely on farmers' experience or consultation with agricultural experts. While experienced farmers may identify common diseases, rare or new diseases often go unnoticed or are misdiagnosed.

Another issue with traditional methods is the lack of accessibility to expert knowledge in rural areas. Many farmers do not have access to agricultural specialists or extension services. Even when experts are available, manual inspection is time-consuming and may not be feasible for large-scale farming.

To address these issues, modern technology is being introduced into agriculture. Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) are transforming the agricultural sector by providing automated solutions. Computer vision techniques enable machines to analyze images of crops and detect diseases with high accuracy. These systems can identify patterns and features that may not be visible to the human eye.

In recent years, large multimodal models (LMMs) have gained popularity. These models are capable of processing multiple types of data, such as images and text, simultaneously. This makes them highly effective for applications like crop disease detection, where both visual analysis and contextual understanding are important.

Agri-LLaVA is one such multimodal model designed for agricultural applications. It can analyze crop images and generate descriptive outputs, including disease identification and suggestions. However, despite its advanced capabilities, it suffers from certain limitations. One major issue is hallucination, where the model generates incorrect or misleading information. This can be harmful in agricultural applications, where decisions directly impact crop health and farmer income.

Another limitation is the dependency on training data. If the dataset is limited, imbalanced, or lacks diversity, the model may not perform well in real-world conditions. For example, a model trained only on specific crops may fail to identify diseases in other crops.

To overcome these challenges, this project proposes an enhanced agricultural assistant that integrates Retrieval-Augmented Generation (RAG). This approach allows the model to retrieve relevant information from reliable sources before generating responses. By grounding the output in verified knowledge, the system reduces hallucination and improves accuracy.

The project also focuses on improving dataset quality by including diverse and real-world images. This ensures that the model can handle different crops, diseases, and environmental conditions effectively. Overall, the topic of this project is centered on developing a smart and reliable AI-based system for crop disease detection and management.

1.2 Brief Introduction of the Title

The title of this project is **“RAG Enhanced Global Assistant for Agriculture Pest and Disease Management”**. This title reflects the key components and objectives of the system.

The term “Enhanced Agricultural Assistant” indicates that the system is designed to assist farmers and agricultural experts in identifying crop diseases and providing recommendations. It is enhanced because it improves upon existing systems by addressing their limitations.

“Retrieval-Augmented Generation (RAG)” is a technique that combines information retrieval with text generation. In this approach, the system first retrieves relevant information from external knowledge sources and then uses that information to generate accurate responses. This ensures that the output is based on real data rather than assumptions.

“Crop Disease Detection” refers to the process of identifying diseases affecting plants using image analysis and AI techniques. Early detection is important to prevent the spread of diseases and minimize crop losses.

The system works by taking an image of a crop as input. The image is processed using a trained model to extract features and identify patterns. The model then retrieves relevant information from a knowledge base, such as agricultural databases or research articles. Finally, it generates a response that includes disease identification and recommendations for treatment and prevention.

This project also emphasizes dataset diversity. A large and varied dataset is used to train the model, including images from different crops, regions, and environmental conditions. This helps the model generalize better and perform well in real-world scenarios.

In addition to detection, the system provides actionable insights. For example, it may suggest the use of specific pesticides, preventive measures, or farming practices. This makes the system more useful and practical for farmers.

Overall, the title represents a comprehensive solution that combines AI, data, and knowledge to improve agricultural outcomes.

1.3 Aim of the Project

The main aim of this project is to develop an intelligent, efficient, and reliable agricultural assistant that can accurately detect crop diseases and provide useful recommendations using advanced Artificial Intelligence techniques. The system is designed to support farmers and agricultural experts by offering a smart solution for early disease identification, which plays a crucial role in improving crop health and productivity.

Agriculture is highly dependent on timely decision-making, and one of the major challenges faced by farmers is the accurate identification of crop diseases. Traditional methods of disease detection rely on manual observation and expert consultation, which can be time-consuming, costly, and sometimes inaccurate. Therefore, this project aims to automate the process of disease detection using modern AI-based approaches, enabling faster and more precise results.

The proposed system focuses on analyzing crop images using deep learning and computer vision techniques. By identifying patterns such as discoloration, spots, and texture variations, the system can detect diseases at an early stage. Early detection is essential in preventing the spread of diseases and minimizing crop losses. This contributes directly to improving agricultural productivity and ensuring better quality yields.

The specific objectives of the project are as follows:

- To design and develop a system that can analyze crop images and identify diseases accurately using advanced AI models
- To reduce hallucination issues in AI models by integrating Retrieval-Augmented Generation, ensuring that outputs are based on reliable and verified knowledge
- To improve the overall reliability and trustworthiness of the system, making it suitable for real-world agricultural applications
- To enhance dataset quality by incorporating diverse and real-world agricultural images, enabling better model generalization
- To provide actionable recommendations for disease treatment and prevention, helping farmers take appropriate measures
- To support farmers in making informed decisions regarding crop management and disease control
- To reduce crop losses caused by delayed or incorrect disease identification
- To improve overall agricultural productivity and efficiency
- To promote sustainable agricultural practices by encouraging the use of appropriate and eco-friendly solutions

Another important aim of this project is to bridge the gap between modern technology and traditional farming practices. Many farmers, especially in rural areas, do not have access to

advanced tools or expert guidance. By developing an easy-to-use system, this project enables farmers to benefit from AI technology without requiring technical expertise. The system can be further extended into mobile or web-based applications, making it accessible to a wider audience.

Scalability is also a key focus of this project. The system is designed to handle large datasets and support multiple crop types. It should be capable of adapting to different environmental conditions, regions, and agricultural practices. This ensures that the system remains useful and effective across various scenarios.

In addition, the project aims to integrate advanced techniques such as Retrieval-Augmented Generation, which enhances the accuracy of predictions by incorporating external knowledge sources. This approach ensures that the system provides reliable and up-to-date information, reducing the risk of incorrect recommendations.

In the long term, this project aims to contribute to global food security by improving crop health and reducing losses. By enabling early detection and effective management of crop diseases, the system supports sustainable agriculture and helps meet the growing demand for food. Furthermore, the project opens opportunities for integration with other technologies such as the Internet of Things (IoT), cloud computing, and smart farming systems.

Overall, the aim of this project is not only to develop a technical solution but also to create a practical and impactful system that can support farmers, improve agricultural outcomes, and contribute to the advancement of modern agriculture.

1.4 Scope of the Project

The scope of this project includes the design, development, and implementation of an Artificial Intelligence (AI)-based system for crop disease detection and recommendation. The system aims to provide an efficient and reliable solution for identifying plant diseases at an early stage and assisting farmers in taking appropriate actions to improve crop health and productivity. By integrating advanced technologies such as deep learning, computer vision, and Retrieval-Augmented Generation (RAG), the project addresses key challenges in modern agriculture.

The proposed system is designed to perform multiple functions that work together to achieve accurate disease detection and effective decision support. One of the primary functions of the system is **image processing and feature extraction**. The system accepts images of crops as input, which may be captured using smartphones or cameras. These images are preprocessed to improve quality by removing noise, adjusting brightness, and resizing them for model compatibility. Feature extraction techniques are then applied to identify important characteristics such as color variations, texture patterns, and structural changes in leaves or plants.

Another important function within the scope of the project is **disease identification using AI models**. The extracted features are analyzed using deep learning models, such as convolutional neural networks (CNNs) or multimodal models, to classify the disease affecting the crop. The system is trained using a diverse dataset, enabling it to recognize different types of diseases across various crops. This automated approach reduces the dependency on manual inspection and increases the speed and accuracy of diagnosis.

The system also incorporates **retrieval of relevant information from knowledge sources**, which is a key enhancement over traditional models. Using Retrieval-Augmented Generation, the system retrieves relevant data from external agricultural knowledge bases, such as research articles, databases, and expert guidelines. This ensures that the system's predictions are supported by verified information, improving reliability and reducing the chances of incorrect outputs.

Another significant aspect of the system is the **generation of accurate and reliable outputs**. The system combines the results of the AI model with retrieved knowledge to produce a final output that includes disease identification and detailed explanation. This helps users understand the condition of the crop and the reasons behind the diagnosis.

In addition to detection, the system provides **recommendations for treatment and prevention methods**. These recommendations may include the use of specific pesticides, organic solutions, preventive measures, and best agricultural practices. This feature makes the system more practical and useful for farmers, as it not only identifies problems but also suggests solutions.

The project has several **practical applications** in the field of agriculture. It can be used in smart farming systems to continuously monitor crop health and detect diseases at an early stage. The

system can also be integrated into mobile applications, allowing farmers to capture images of crops and receive instant feedback directly in the field. This improves accessibility and enables farmers to make quick decisions. Additionally, agricultural experts and researchers can use the system for analysis, study, and development of improved farming techniques.

The scope of the project can be further extended in several ways. One possible extension is the **development of a mobile or web-based application**, which would make the system more accessible to users. Another extension is the **integration with Internet of Things (IoT) devices**, such as sensors that monitor environmental conditions like temperature, humidity, and soil moisture. Combining this data with image analysis can improve the accuracy of disease detection.

The system can also be enhanced by **expanding the dataset** to include a wider range of crops, diseases, and environmental conditions. This would improve the model's ability to generalize and perform well in different scenarios. Additionally, the system can be extended to support **multiple languages**, making it more accessible to farmers from different regions and backgrounds.

Another important extension is **cloud-based deployment**, which allows the system to handle large-scale data and support multiple users simultaneously. Cloud integration also enables continuous updates and improvements to the system without requiring local installation.

Despite its advantages, the project has certain **limitations**. The performance of the system depends heavily on the quality and diversity of the dataset used for training. If the dataset is limited or biased, the model may not perform well in real-world conditions. Additionally, training deep learning models requires significant computational resources, which may not be readily available in all environments. The retrieval mechanism may also require internet connectivity to access external knowledge sources, which can be a challenge in remote areas.

However, these limitations can be addressed in future work through improved data collection, optimization techniques, and infrastructure development. Overall, the project provides a strong foundation for the development of intelligent agricultural systems.

In conclusion, the scope of this project covers a wide range of functionalities and applications, making it a comprehensive solution for crop disease detection and management. By combining

advanced AI techniques with practical features, the system has the potential to significantly improve agricultural productivity and support sustainable farming practices.

1.5 Organization of the thesis

The thesis is systematically organized to present the complete development and analysis of the proposed project in a clear and structured manner. Each part of the document is arranged in a logical sequence so that the reader can easily understand the background, methodology, implementation, and outcomes of the work. The organization ensures a smooth flow of information, starting from the introduction of the problem to the final conclusions and future directions.

The initial part of the thesis focuses on providing a comprehensive overview of the project. It introduces the problem domain, highlights the importance of crop disease detection in agriculture, and explains the motivation behind the proposed work. This section also outlines the key objectives and scope of the project, helping the reader understand the purpose and significance of the study.

Following this, the thesis presents a detailed review of existing research and technologies related to crop disease detection and Artificial Intelligence. Various approaches, models, and methodologies are discussed to provide a clear understanding of the current state of the field. The strengths and limitations of these existing systems are analyzed, which helps in identifying the research gaps. This forms the foundation for proposing an improved solution.

The next part of the thesis focuses on the design and development of the proposed system. It explains the overall architecture, working principles, and components of the system in detail. This includes dataset preparation, preprocessing techniques, model selection, and the integration of advanced concepts such as Retrieval-Augmented Generation. The explanation is structured in a way that clearly describes how the system processes input data and generates accurate outputs.

Subsequently, the thesis presents the results obtained from the implementation of the proposed system. The results are analyzed to understand the effectiveness of the system and its ability to perform under different conditions. Comparisons with existing approaches are also discussed to demonstrate the improvements achieved by the proposed method.

The final part of the thesis provides a summary of the entire work, highlighting the key findings and contributions of the project. It discusses how the proposed system addresses the identified research gaps and improves upon existing solutions. In addition, this section outlines possible directions for future work, including enhancements, extensions, and potential real-world applications.

Overall, the thesis is organized in a coherent and systematic manner, ensuring that each section contributes to a complete understanding of the project. This structured approach not only improves readability but also helps in effectively presenting the technical and practical aspects of the work.

LITERATURE REVIEW

2.1 Existing system analysis

Paper 1: InstructBLIP (2024)

This paper introduces InstructBLIP, a vision-language model designed for general-purpose tasks using instruction tuning. It improves understanding of both images and text by aligning them through instructions. The model performs well in various tasks like captioning and question answering. However, it still faces challenges in domain-specific applications like agriculture due to lack of specialized training data.

Paper 2: LLaVA-Med (2024)

This paper presents LLaVA-Med, a biomedical vision-language assistant trained efficiently in a short time. It shows strong performance in medical image understanding and reasoning tasks. The system is useful in healthcare applications but is limited to the biomedical domain and cannot generalize well to agriculture-related problems.

Paper 3: Mini-Gemini (2024)

Mini-Gemini focuses on improving multimodal models by optimizing performance with fewer resources. It enhances efficiency and reduces computational cost. While it performs well in multimodal tasks, it still lacks robustness in handling real-world agricultural datasets.

Paper 4: Improved Baselines with Visual Instruction Tuning (2024)

This work improves existing models using visual instruction tuning techniques. It enhances performance in vision-language tasks by better aligning visual features with textual instructions. However, the approach depends heavily on high-quality annotated data.

Paper 5: Visual Instruction Tuning (2024)

This paper introduces visual instruction tuning for improving multimodal models. It allows models to follow instructions more effectively when processing images. Although it improves general performance, it does not address hallucination issues.

Paper 6: ALLaVA (2024)

ALLaVA uses GPT-4V-generated data to train a lightweight vision-language model. It reduces training cost and improves efficiency. However, reliance on synthetic data may introduce bias and reduce accuracy in real-world applications.

Paper 7: Efficient Vision-Language Instruction Tuning (2024)

This paper proposes a cost-effective approach for training vision-language models. It reduces training time and computational requirements. Despite efficiency, it may compromise performance in complex scenarios.

Paper 8: VisionLLM (2024)

VisionLLM extends language models to handle vision-based tasks. It acts as a decoder for visual inputs and performs well in open-ended tasks. However, it requires large-scale data and resources.

Paper 9: Qwen-VL (2023)

Qwen-VL is a versatile model capable of image understanding, localization, and text recognition. It performs multiple tasks effectively. However, it may generate incorrect outputs due to hallucination.

Paper 10: GPT-4 (AGI Study) (2023)

This paper explores the capabilities of GPT-4 in achieving artificial general intelligence. It demonstrates strong reasoning and multimodal abilities. However, it is not specifically designed for agriculture applications.

Paper 11: X-LLM (2023)

X-LLM treats multimodal inputs as different languages. It improves model understanding across modalities. However, it requires complex training and large datasets.

Paper 12: ShareGPT4V (2023)

This paper improves multimodal models using better caption datasets. It enhances image-text alignment. However, performance depends heavily on dataset quality.

Paper 13: PaLM (2023)

PaLM is a large-scale language model that improves performance through scaling. It achieves strong results in NLP tasks. However, it lacks direct support for vision-based tasks.

Paper 14: LLaMA Adapter V2 (2023)

This paper introduces a parameter-efficient approach to adapt LLaMA for vision tasks. It reduces training cost while maintaining performance. However, it may not fully capture complex visual features.

Paper 15: Multimodal GPT (2023)

This model integrates vision and language for conversational tasks. It allows interaction with users using images and text. However, it may struggle with domain-specific accuracy.

Paper 16: LISA (2023)

LISA focuses on reasoning-based segmentation using LLMs. It improves understanding of visual content. However, it is mainly focused on segmentation tasks.

Paper 17: BLOOM (2023)

BLOOM is a multilingual language model supporting multiple languages. It improves accessibility and inclusivity. However, it does not directly handle image data.

Paper 18: Macaw-LLM (2023)

This model integrates multiple modalities including image, audio, and video. It supports diverse inputs but requires high computational power.

Paper 19: Video-ChatGPT (2023)

This paper extends multimodal models to video understanding. It performs well in analyzing video content but is not optimized for static agricultural images.

Paper 20: GPT-4V System Card (2023)

This document explains the capabilities of GPT-4 Vision. It shows strong multimodal reasoning but highlights issues like hallucination and bias.

Paper 21: DetGPT (2023)

DetGPT focuses on object detection using reasoning. It improves detection accuracy but is limited to specific tasks.

Paper 22: PandaGPT (2023)

PandaGPT is a multimodal model supporting multiple input types. It improves flexibility but requires large training data.

Paper 23: Stanford Alpaca (2023)

Alpaca is an instruction-following model derived from LLaMA. It is cost-efficient but has limited multimodal capabilities.

Paper 24: LLaMA (2023)

LLaMA is a foundational language model. It provides strong NLP performance but lacks multimodal features.

Paper 25: GPT-4V Prompting (2023)

This paper improves visual instruction tuning using better prompts. It enhances performance but still suffers from hallucination.

Paper 26: CogVLM (2023)

CogVLM enhances visual understanding in language models. It improves reasoning but requires high resources.

Paper 27: mPLUG-Owl (2023)

This model enables modular multimodal learning. It improves flexibility but adds system complexity.

Paper 28: MultiInstruct (2022)

This paper improves zero-shot learning using instruction tuning. It enhances generalization but lacks domain-specific accuracy.

Paper 29: Flamingo (2022)

Flamingo is a vision-language model designed for few-shot learning. It performs well with limited data but requires large pretraining.

Paper 30: Wheat Disease Detection (2021)

This paper uses CNN for detecting wheat diseases. It achieves good accuracy but is limited to specific crops.

Paper 31: Zero-shot Learners (2021)

This work shows that fine-tuned models can perform zero-shot tasks. However, accuracy may vary across domains.

Paper 32: Few-shot Learners (2020)

This paper demonstrates few-shot learning capabilities in language models. It reduces training needs but may lack precision.

Paper 33: IP102 Dataset (2019)

This dataset provides large-scale insect pest images. It supports research but lacks diversity in crops.

Paper 34: Plant Disease Dataset (2015)

This paper introduces an open dataset for plant disease detection. It is widely used but limited in real-world variation.

Paper 35: Evaluation Metrics (2005)

This paper explains precision, recall, and F-score. These metrics are important for evaluating model performance.

Sr No.	Title of Paper	Author	Year	Methodology	Dataset	Research Gap	Performance
1	Allava: Harnessing gpt4v-synthesized data	Guiming Hardy Chen, Shunian Chen, Ruifei Zhang, et al.	2024	Synthetic GPT-4V captions for training lightweight models	LAION, Vision FLAN	Relies heavily on synthetic data which may contain hallucinations.	Efficient with small compute
2	Instructblip: Towards general-purpose vision-language models	Wenliang Dai, Junnan Li, Dongxu Li, Anthony Meng, et al.	2024	Instruction-aware Query Transformer	26 public VL datasets	General instruction following is good, but fails in subtle disease differentiation.	Strong instruction following
3	Llava-med: Training a large language-and-vision assistant	Chunyuan Li, Cliff Wong, Sheng Zhang, Naoto Usuyama, et al	2024	Fine-tuning on biomedical data (PubMed)	PubMed Vision	Medical focus; proves domain tuning works but not directly usable for crops.	Improved reasoning

4	Mini-gemini: Mining the potential of multi-modality	Yanwei Li, Yuechen Zhang, Chengyao Wang, et al.	2024	Dual vision encoders for high-resemblances	Synthetic multimodal data	High resolution is good, but requires specific agri-data alignment.	Zero-shot benchmarks
5	Improved Baselines with Visual Instruction Tuning	Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee	2024	LLaVA improved with MLP connector	COCO, VQA	The baseline for Agri-LLaVA, but lacks external knowledge retrieval.	Improved efficiency
6	Visual Instruction Tuning (LLaVA)	Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee	2024	GPT-4 generates multimodal instructions	COCO	Hallucinates facts when domain knowledge is missing.	85.1% on synthetic data
7	Efficient vision language instruction tuning	Gen Luo, Yiyi Zhou, Tianhe Ren, Shengxin Chen, et al.	2024	MMA connects encoders and LLMs via adapters	COCO, VQAv2	Focus on efficiency over depth of knowledge.	Low training cost
8	Visionllm: Large language model is also an open-ended decoder	Wenhai Wang, Zhe Chen, Xiaokang Chen, Jiannan Wu, et al.	2024	Unified vision decoder model	Visual benchmarks	Good for segmentation, but lacks the reasoning depth for complex biology.	General visual task solver
9	Qwen-vl: A versatile vision-language model	J Bai, S Bai, S Yang, S Wang, et al.	2023	Versatile vision-language understanding backbone	COCO, LVIS	Not optimized for biological taxonomy or symptom recognition.	Robust multitask ability
10	Sparks of AGI: Early experiments with gpt-4	Sebastien Bubeck, Varun Chandrasekar, et al.	2023	Evaluations across reasoning and vision	Proprietary / Not Disclosed	Closed source; hallucinations in specialized scientific fields.	Early AGI behaviors

11	X-llm: Bootstrapping advanced LLMs	Feilong Chen, Minglun Han, Haozhi Zhao, et al.	2023	Multimodal pretraining via text-language mapping	Specific Dataset Not Disclosed	Treats modalities as foreign languages; complex to adapt for pest data.	Flexible cross-modality transfer
12	Sharegpt4v: Improving large multi-modal models	Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, et al.	2023	Enhanced caption datasets for fine-tuning	COCO, LAION, CC3M	General captions lack the scientific precision needed for agronomy.	Better caption alignment
13	PaLM: Scaling language modeling with pathways	Aakanksha Chowdhery, Sharan Narang, Jacob Devlin, et al.	2023	Transformer-based, large distributed training	Massive web corpus	Unimodal (text-only focus in base); requires massive compute resources.	Strong reasoning
14	Llama-adapter v2: Parameter-efficient visual instruction model	Peng Gao, Jiaming Han, Renrui Zhang, Ziyi Lin, et al.	2023	Parameter Efficient visual instruction tuning	COCO Caption	Lightweight but may lack depth for complex biological reasoning.	Easy domain adaptation
15	Multimodal-gpt: A vision and language model for dialogue	Tao Gong, Chengqi Lyu, Shilong Zhang, Yudong Wang, et al.	2023	GPT backbone + image encoder	VQA v2.0, OKVQA	Struggles with fine-grained visual details necessary for early pest detection.	Visual-aware chat
16	Lisa: Reasoning segmentation via large language model	Xin Lai, Zhuotao Tian, Yukang Chen, Yanwei Li, et al.	2023	Embedding-as-mask to enable segmentation	COCO, ADE20K	Focuses on segmentation, not biological diagnosis or treatment advice.	Strong zero-shot segmentation
17	Bloom: A 176b-parameter open-access model	Teven Le Scao, Angela Fan, Christopher Akiki, et al.	2023	Transformer trained on multilingual corpora	ROOTS corpus	Text-only; requires adaptation for visual inputs.	Competitive performance

18	Macaw-llm: Multi-modal language	Chenyang Lyu, Minghao Wu, Longyue Wang, et al	2023	Cross-modal fusion (Image, Audio, Video)	Multi-modal datasets	Integration of video is promising but complex for simple diagnostic apps.	Multi-sensory understanding
19	Video-chatgpt: Towards detailed video understanding	Muhammad Maaz, Hanoona Rasheed, Salman Khan, et al.	2023	Frame encoder + GPT backbone	VideoQA datasets	Video focused; higher compute latency for real-time field use.	Detailed temporal reasoning
20	GPT-4 (Vision) System Card	OpenAI	2023	Multimodal architecture	Internal OpenAI data	Closed source; API costs are prohibitive for small farmers.	Powerful visual reasoning
21	Detgpt: Detect what you need via reasoning	Renjie Pi, Jiahui Gao, Shizhe Diao, Rui Pan, et al.	2023	Visual reasoning + LLM	Detection datasets	Good for locating pests, but weak on treatment recommendation.	Detects objects by reasoning
22	Pandagpt: One model to instruction-follow them all	Yixuan Su, Tian Lan, Huayang Li, Jialu Xu, et al.	2023	Multimodal instruction alignment	Visual-language data	General-purpose instruction following; lacks specialized agricultural ontology.	Universal instruction model
23	Stanford alpaca: An instruction-following llama model	Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, et al.	2023	Fine-tuned LLaMA with self-instruct data	Text instruction data	Text-only model; requires integration with vision encoders for crop diagnosis.	Strong alignment
24	Open and efficient foundation language models	Hugo Touvron, Thibaut Lavril, Gautier Izacard, et al.	2023	Optimized transformer training	Web, books, code	Base foundation model; no visual capabilities or pest knowledge.	Strong open model

25	To see is to believe: Prompting gpt-4v	Junke Wang, Lingchen Meng, Zejia Weng, Bo He, et al.	2023	Prompt engineering study for visual tasks	GPT-4V image tasks	Focuses on prompting techniques rather than architectural changes.	Improved visual responses
26	Cogvlm: Visual expert for pretrained language models	Weihan Wang, Qingsong Lv, Wenmeng Yu, Wenyi Hong, et al.	2023	Vision-aware fine-tuning	Image-caption datasets	High computational cost; general visual expert may miss subtle disease patterns	Strong visual understanding
27	Flamingo: A visual language model for few-shot learning	Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, et al.	2022	Visual encoder + frozen LM with cross-attention	Multi Modal Massive Web, ALIGN	Lacks specific fine-tuning for agricultural nuances (pests/diseases).	Strong few-shot reasoning
28	Finetuned language models are zero-shot learners	Jason Wei, Maarten Bosma, Vincent Y Zhao, et al.	2021	Fine-tuned GPT models via tuning	Text corpora	Zero-shot learning is often insufficient for high-stakes crop identification.	Improves zero-shot learning
29	Leaf and spike wheat disease detection	Lakshay Goyal, Chandra Mani Sharma, Anupam Singh, et al.	2021	Deep CNN for classification with SVM	LWDCD2020	Classification only; no conversational or explanation capability.	97.88% accuracy
30	Language models are few-shot learners	Ben Mann, N Ryder, M Subbiah, J Kaplan, et al.	2020	Transformer pretraining (GPT-3)	WebText, Books	Text only; outdated compared to GPT-4 based architectures.	Few-shot capabilities
31	Ip102: A large-scale benchmark dataset	Xiaoping Wu, Chi Zhan, Yu-Kun Lai, Ming-Ming Cheng, et al.	2019	Insect pest recognition benchmark	102 insect classes	A dataset paper, not a solution model; data is lab-based.	Benchmark accuracy >90%

32	An open access repository of images on plant health	David Hughes, Marcel Salathe, et al.	2015	Data collection and curation	Plant Village	Dataset is lab-controlled; lacks background complexity of real fields.	Core training source
33	A probabilistic interpretation of precision recall and f-score	Cyril Goutte and Eric Gaussier	2005	Probabilistic evaluation theory	Benchmark IR datasets	Theoretical metric paper; not a model implementation.	Better metric interpretation

Table 2.1 Literature Survey

2.2 Research Gap

The rapid development of Artificial Intelligence, particularly in the fields of deep learning and multimodal models, has significantly improved the capability of automated systems in various domains, including agriculture. However, despite these advancements, several limitations still exist in current approaches for crop disease detection and agricultural assistance. These limitations highlight the need for further research and development.

One of the most critical gaps identified in existing systems is the problem of **hallucination in large multimodal models**. Models such as vision-language systems are capable of generating descriptive outputs based on image and text inputs. However, these outputs are often not grounded in verified knowledge, leading to incorrect or misleading information. In the context of agriculture, such inaccuracies can result in improper treatment recommendations, which may further damage crops and reduce yield. Therefore, there is a strong need for systems that can ensure factual correctness and reliability.

Another significant issue is the **lack of diverse and representative datasets**. Most existing models are trained on datasets that are limited in size and diversity. These datasets often consist of images captured under controlled conditions with uniform backgrounds and lighting. In real-world scenarios, crops are exposed to varying environmental conditions such as different lighting, weather, and soil types. As a result, models trained on limited datasets fail to generalize effectively. This creates a gap between research performance and real-world applicability.

Furthermore, many existing systems focus primarily on **disease classification** and do not provide complete solutions. While identifying the disease is important, farmers also require actionable insights such as treatment methods, preventive measures, and best practices for crop management. The absence of such recommendations reduces the usefulness of these systems in practical applications.

Another important gap is the **lack of integration with external knowledge sources**. Most AI models rely solely on pre-trained knowledge and do not update their understanding dynamically. This is a major limitation, especially in agriculture, where new diseases and treatment methods are continuously emerging. Without access to updated information, models may provide outdated or irrelevant responses.

The issue of **computational complexity and resource requirements** is also a major concern. Many advanced models require high computational power, making them difficult to deploy in rural or resource-constrained environments. Farmers in such areas may not have access to high-end devices or stable internet connections, limiting the usability of these systems.

Additionally, there is a lack of **user-friendly interfaces** and accessibility features in existing systems. Most research models are designed for technical users and do not consider the needs of farmers, who may not have technical knowledge. This creates a barrier to adoption and limits the impact of these technologies.

Another gap is the **lack of multilingual support**, which is essential in countries like India where multiple languages are spoken. Systems that do not support local languages may not be easily usable by farmers.

Moreover, there is limited focus on **handling rare and complex diseases**. Many datasets and models focus on common diseases, leaving rare cases underrepresented. This reduces the reliability of the system in diverse agricultural conditions.

In conclusion, the major research gaps identified are:

- Lack of grounding mechanisms to reduce hallucination
- Limited and non-diverse datasets

- Absence of recommendation systems
- Lack of integration with real-time knowledge bases
- High computational requirements
- Limited usability and accessibility
- Lack of multilingual support
- Poor handling of rare and complex cases

These gaps emphasize the need for an advanced system that can overcome these challenges and provide a reliable solution for agricultural applications.

2.3 Proposed system

To address the limitations identified in existing systems, this project proposes an **Enhanced Agricultural Assistant** that integrates **multimodal learning with Retrieval-Augmented Generation (RAG)**. The proposed system aims to provide accurate, reliable, and practical solutions for crop disease detection and management.

The system is designed to function as an intelligent assistant that can analyze crop images, identify diseases, and provide actionable recommendations. It combines the strengths of deep learning, multimodal models, and knowledge retrieval techniques to achieve improved performance.

2.3.1 System Overview

The proposed system consists of multiple components that work together to process input data and generate outputs. These components include image processing modules, feature extraction layers, multimodal models, retrieval mechanisms, and response generation units.

The system begins with the acquisition of input data, which is typically an image of a crop captured using a smartphone or camera. The image is then processed to remove noise and enhance quality. Feature extraction techniques are applied to identify important visual patterns such as color changes, spots, and texture variations.

2.3.2 Multimodal Learning Approach

The system uses a multimodal model that can process both visual and textual information. This allows the model to not only analyze images but also understand contextual information related to crop diseases. By combining these two types of data, the system can generate more accurate and meaningful outputs.

2.3.3 Retrieval-Augmented Generation (RAG)

One of the key features of the proposed system is the integration of Retrieval-Augmented Generation. In this approach, the system retrieves relevant information from external knowledge sources before generating a response. These sources may include agricultural databases, research articles, and expert guidelines.

The retrieved information is then combined with the model's prediction to produce a final output. This ensures that the response is grounded in verified knowledge, reducing the chances of hallucination.

2.3.4 Dataset Enhancement

The system uses an enhanced dataset that includes images from multiple sources. The dataset covers various crops, diseases, pests, and environmental conditions. This diversity improves the model's ability to generalize and perform well in real-world scenarios.

2.3.5 Recommendation System

In addition to disease detection, the system provides recommendations for treatment and prevention. These recommendations are based on retrieved knowledge and may include:

- Use of pesticides
- Organic treatment methods
- Preventive measures
- Crop management practices

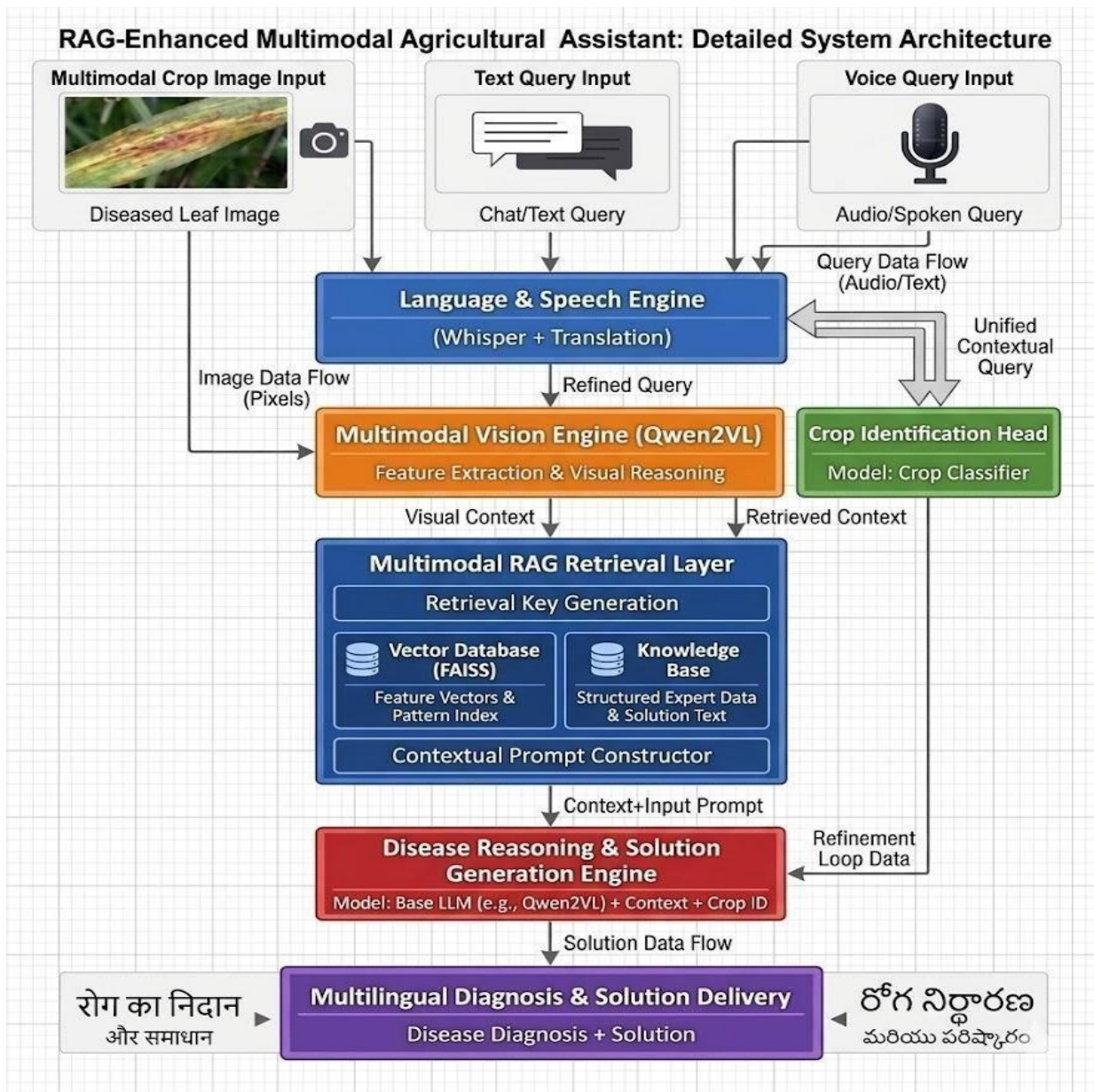


Fig. 2.1 Architecture Diagram

2.3.6 Advantages of Proposed System

- Improved accuracy and reliability
- Reduced hallucination
- Better generalization
- Practical recommendations
- Scalability and flexibility

2.4 Objectives of the Project

The objectives of the project are defined to guide the design, development, and evaluation of the proposed system. These objectives focus on improving the accuracy, reliability, and usability of crop disease detection systems using advanced Artificial Intelligence techniques. Each objective addresses specific challenges identified in existing systems and contributes to building a robust and practical agricultural assistant.

1. To develop an intelligent system for crop disease detection using Artificial Intelligence

The primary objective of this project is to design and develop an intelligent system capable of accurately detecting crop diseases from images. The system utilizes advanced Artificial Intelligence techniques, particularly deep learning and computer vision, to analyze visual features such as color variations, texture patterns, and leaf structure.

Traditional methods of disease detection rely on manual observation, which can be time-consuming and prone to errors. By automating this process, the proposed system aims to provide faster and more accurate results. The system is trained using a dataset of crop images, enabling it to learn patterns associated with different diseases. This objective ensures that the system can perform reliable disease identification even in complex scenarios.

2. To integrate Retrieval-Augmented Generation (RAG) for improving reliability

Another key objective of this project is to enhance the reliability of the system by integrating Retrieval-Augmented Generation (RAG). One of the major limitations of existing AI models is their tendency to generate incorrect or misleading information, commonly known as hallucination.

To address this issue, the proposed system incorporates a retrieval mechanism that accesses external knowledge sources such as agricultural databases and research articles. Before generating a response, the system retrieves relevant information and uses it to support its predictions. This ensures that the output is grounded in verified knowledge, thereby improving accuracy and trustworthiness.

This objective plays a crucial role in making the system suitable for real-world agricultural applications, where reliable information is essential for decision-making.

3. To enhance dataset diversity for better generalization

The performance of any AI model largely depends on the quality and diversity of the dataset used for training. Therefore, another important objective of this project is to enhance the dataset by including images from various sources, crops, diseases, and environmental conditions.

Many existing systems are trained on limited datasets that do not represent real-world conditions. As a result, they fail to generalize effectively when deployed in practical scenarios. By incorporating diverse data, the proposed system aims to improve its robustness and adaptability.

This objective ensures that the model can accurately detect diseases under different lighting conditions, backgrounds, and crop variations, making it more reliable in real-world applications.

4. To provide actionable recommendations for farmers

In addition to disease detection, the system is designed to provide actionable recommendations for farmers. Identifying a disease alone is not sufficient; farmers also need guidance on how to treat and prevent it.

The proposed system generates recommendations based on retrieved knowledge, including:

- Suitable pesticides or treatments
- Organic and eco-friendly solutions
- Preventive measures
- Best agricultural practices

This objective enhances the practical usability of the system and helps farmers make informed decisions. It also contributes to sustainable agriculture by promoting efficient and responsible farming practices.

5. To design a scalable and user-friendly system

The final objective of the project is to develop a system that is both scalable and easy to use. Scalability ensures that the system can handle large datasets, support multiple crops, and adapt to future requirements. The system architecture is designed in a modular way, allowing new features and data to be added without major changes.

User-friendliness is equally important, as the system is intended for farmers who may not have technical expertise. The interface should be simple and intuitive, allowing users to interact with the system. In future, the system can be extended into a mobile or web application for easy access.

This objective ensures that the system is practical, accessible, and capable of being widely adopted in real-world agricultural environments.

2.5 Scope of the Present Work

The scope of the present work focuses on the design, development, and evaluation of an intelligent agricultural assistant that leverages advanced Artificial Intelligence techniques for crop disease detection and management. The system aims to provide a comprehensive solution that not only identifies diseases but also assists farmers in taking appropriate actions to improve crop health and productivity.

The proposed system primarily deals with **image-based crop disease detection**, where the input is an image of a plant or crop affected by disease. The system processes this image using deep learning and multimodal models to identify patterns and symptoms associated with different diseases. This approach enables early detection, which is crucial in preventing the spread of diseases and minimizing crop losses.

A key aspect within the scope of this project is the integration of **multimodal learning**, which allows the system to process both visual and textual information. Unlike traditional systems that rely only on images, this approach enables the model to understand contextual information and provide more meaningful outputs. The inclusion of textual reasoning enhances the system's capability to interpret complex disease patterns and provide detailed explanations.

Another important component of the project scope is the use of **Retrieval-Augmented Generation (RAG)**. This technique enables the system to retrieve relevant information from external knowledge bases before generating a response. By grounding the output in verified data, the system significantly reduces the chances of hallucination and improves the reliability of its predictions. This makes the system more suitable for real-world agricultural applications, where accuracy is critical.

The project also emphasizes the importance of **dataset diversity and quality**. The system is trained using a dataset that includes images from various crops, diseases, and environmental conditions. This ensures that the model can generalize effectively and perform well in different scenarios. The dataset may include images captured under varying lighting conditions, backgrounds, and resolutions, making the system robust and adaptable.

Within the scope of this work, the system is also designed to provide **actionable recommendations**. After identifying the disease, the system suggests appropriate treatment methods, preventive measures, and best practices for crop management. These recommendations are based on retrieved knowledge and are aimed at helping farmers make informed decisions. This feature enhances the practical usability of the system.

The scope of the project extends to the development of a **user-friendly interface**, which allows users to interact with the system easily. Although the current implementation may focus on a prototype, the system can be extended into a mobile or web-based application. This would enable farmers to use the system directly in the field by capturing images using smartphones.

Another important aspect is the **scalability of the system**. The proposed solution is designed in such a way that it can be extended to support multiple crops, diseases, and regions. The system architecture allows for the addition of new data and features without significant changes to the existing framework. This ensures that the system remains relevant and adaptable to future requirements.

The project also considers the potential integration of the system with **emerging technologies** such as the Internet of Things (IoT). For example, sensors can be used to collect real-time data on environmental conditions, which can be combined with image data to improve disease detection

accuracy. Although this integration is not fully implemented in the current work, it represents a possible direction for future development.

Furthermore, the system can be extended to include **multilingual support**, allowing users to interact with the system in different languages. This is particularly important in regions where farmers may not be comfortable using English. By providing support for local languages, the system can become more accessible and widely adopted.

The scope of the project also includes **performance evaluation and validation**. The system is tested using various metrics such as accuracy, precision, recall, and F1-score. These metrics help in assessing the effectiveness of the model and comparing it with existing approaches. The evaluation ensures that the system meets the required performance standards.

However, the scope of the present work is subject to certain limitations. The system's performance depends on the quality and availability of training data. In addition, the retrieval mechanism may require internet connectivity to access external knowledge sources. Computational requirements may also pose challenges in low-resource environments.

Despite these limitations, the project provides a strong foundation for the development of intelligent agricultural systems. The proposed solution can be further enhanced and deployed in real-world scenarios to support farmers and improve agricultural productivity.

In summary, the scope of the present work includes:

- Image-based crop disease detection
- Integration of multimodal learning
- Use of Retrieval-Augmented Generation
- Dataset enhancement and diversification
- Recommendation generation
- User-friendly system design
- Scalability and adaptability
- Potential integration with IoT and multilingual systems
- Performance evaluation and validation

3. METHODOLOGY

3.1 Requirement Analysis

3.1.1 Introduction to Requirement Analysis

Requirement analysis is one of the most important phases in the development of any system. It involves identifying, analyzing, and documenting the resources required to design and implement the proposed solution. In the context of this project, requirement analysis focuses on determining the software tools, hardware configurations, and computational resources necessary to build an intelligent agricultural assistant.

The proposed system involves multiple components such as image processing, multimodal learning, natural language processing, and retrieval-based systems. Each of these components requires specific technologies and tools for efficient functioning. Therefore, careful selection of software and hardware is essential to ensure that the system performs accurately and efficiently.

Requirement analysis also helps in understanding system limitations, optimizing performance, and ensuring compatibility between different components. By defining clear requirements, the development process becomes more structured and manageable.

3.1.2 Software Requirements

The development of the proposed system relies on several software tools, frameworks, and libraries that work together to enable efficient processing of multimodal data, accurate disease detection, and reliable response generation. These tools are carefully selected based on their compatibility, performance, scalability, and ability to support Artificial Intelligence-based applications.

The system integrates components from different domains such as image processing, natural language processing, speech recognition, and information retrieval. Therefore, a combination of modern and well-supported technologies is essential to ensure smooth functioning and high performance.

Operating System: Windows

The proposed system is developed using the Windows operating system, which provides a stable, flexible, and user-friendly environment for software development. Windows is widely used in both academic and industrial settings, making it a practical choice for implementing AI-based applications.

One of the major advantages of using Windows is its compatibility with a wide range of development tools and programming environments. It supports Python-based ecosystems, including virtual environments such as Anaconda, and development platforms like Jupyter Notebook and Visual Studio Code. This makes it easier to manage dependencies, run experiments, and debug code efficiently.

Windows also provides support for GPU acceleration through CUDA and related drivers, which is essential for deep learning tasks. This allows the system to utilize hardware resources effectively, especially during model training and inference.

In addition, the operating system plays a significant role in:

- File handling and storage management
- Process scheduling and execution
- Integration of multiple software components
- Management of system resources

The availability of graphical user interfaces and built-in utilities further simplifies development and testing. Overall, the use of Windows enhances the ease of development, deployment, and maintenance of the system.

Programming Language: Python

Python is selected as the primary programming language for the development of this project due to its simplicity, readability, and extensive support for AI and data science applications. It is one of the most widely used languages in the field of Artificial Intelligence and Machine Learning.

Python offers a rich ecosystem of libraries and frameworks that simplify complex tasks such as image processing, model training, and data analysis. Its syntax is easy to understand, which reduces development time and allows rapid prototyping of ideas.

Key advantages of Python include:

- Simple and readable syntax, making it easy to learn and use
- Large collection of libraries for AI, machine learning, and data processing
- Strong community support and extensive documentation
- Cross-platform compatibility
- Easy integration with external tools and APIs

In this project, Python is used for:

- Data preprocessing and cleaning
- Image processing and feature extraction
- Model implementation and inference
- Integration of RAG components
- Output generation and user interaction

Python also supports object-oriented programming, which helps in designing modular and reusable code. This improves the maintainability and scalability of the system.

NLP Libraries: Hugging Face Transformers

The Hugging Face Transformers library is used for implementing Natural Language Processing functionalities in the system. It provides access to a wide range of pre-trained transformer-based models that can perform advanced language tasks.

Transformers are based on attention mechanisms, which allow models to understand the context and relationships between words in a sentence. This makes them highly effective for tasks such as text generation and understanding.

In this project, the Hugging Face library is used to:

- Process and interpret textual inputs
- Generate meaningful and context-aware responses
- Assist in integrating text with visual information in multimodal models

The use of pre-trained models offers several benefits:

- Reduces the need for training models from scratch
- Saves computational resources and time
- Improves accuracy and performance
- Enables handling of complex language structures

Additionally, the library supports fine-tuning, allowing models to be adapted to specific agricultural datasets and tasks. This enhances the relevance and quality of the generated outputs.

Vision-Language Model: Qwen2-VL

Qwen2-VL is used as the core multimodal model in the proposed system. It is designed to process both visual and textual inputs, making it suitable for applications that require understanding of images along with contextual information.

The model works by encoding image features and combining them with text embeddings to generate outputs. This allows the system to perform tasks such as:

- Identifying crop diseases from images
- Providing descriptions of symptoms
- Generating recommendations based on detected conditions

Key features of Qwen2-VL include:

- Ability to handle multimodal inputs (image + text)
- Strong reasoning and contextual understanding
- High accuracy in visual recognition tasks
- Capability to generate detailed and descriptive outputs

The use of a multimodal model is particularly important in agriculture, where both visual symptoms and contextual information play a role in disease identification. By combining these inputs, the system can produce more accurate and meaningful results.

Database: FAISS (Vector Database)

FAISS (Facebook AI Similarity Search) is used as a vector database to support the retrieval component of the system. It is designed for efficient similarity search and clustering of high-dimensional data.

In the context of this project, FAISS stores embeddings of text and image data. These embeddings represent the semantic meaning of the data in vector form. When a query is made, FAISS retrieves the most relevant information by comparing vector similarities.

Functions of FAISS include:

- Efficient storage of embeddings
- Fast similarity search using distance metrics
- Retrieval of relevant knowledge for response generation

FAISS plays a crucial role in the Retrieval-Augmented Generation process by:

- Enhancing the accuracy of responses
- Reducing hallucination in model outputs
- Providing contextually relevant information

Its ability to handle large datasets efficiently makes it suitable for scalable applications.

Speech Processing: Whisper

Whisper is used for speech-to-text conversion, enabling users to interact with the system using voice input. This feature enhances accessibility, especially for users who may not be comfortable typing.

The speech input provided by the user is converted into text, which is then processed by the system. The output can also be converted back into speech if required.

Advantages of using Whisper include:

- High accuracy in speech recognition
- Support for multiple languages
- Ability to handle noisy audio inputs
- Improved user interaction and accessibility

This feature is particularly useful in agricultural applications, where farmers may prefer voice-based interaction while working in the field.

Translation Tools: FastText and NLLB

FastText and NLLB are used to enable multilingual support in the system. These tools allow the system to communicate with users in different languages, making it more inclusive and accessible.

FastText is used for efficient text representation and classification, while NLLB is designed for high-quality translation across multiple languages.

Functions include:

- Translating user input into a common processing language
- Translating system output into the user's preferred language
- Supporting communication across different linguistic backgrounds

Multilingual support is essential in agricultural systems, especially in countries with diverse languages. It ensures that farmers from different regions can use the system effectively.

Integration of Software Components

All the above software tools are integrated to form a complete system. The interaction between components is as follows:

- Python acts as the central programming environment

- Qwen2-VL processes image and text inputs
- Hugging Face handles language processing
- FAISS retrieves relevant information
- Whisper processes speech input
- FastText and NLLB handle translation

This integration ensures that the system operates smoothly and efficiently, providing accurate and reliable outputs.

In conclusion, the selected software tools and frameworks provide a strong foundation for the development of the proposed system. They enable efficient processing of multimodal data, support advanced AI functionalities, and ensure scalability and flexibility.

The combination of these technologies allows the system to deliver accurate disease detection, reliable recommendations, and user-friendly interaction, making it suitable for real-world agricultural applications.

3.1.3 Hardware Requirements

The hardware requirements for this project are designed to support the computational needs of Artificial Intelligence models, image processing tasks, and retrieval-based systems. Since the proposed system involves deep learning, multimodal processing, and data-intensive operations, it is important to ensure that the hardware configuration is capable of handling these tasks efficiently.

The system is developed in a way that it can operate on standard computing devices; however, performance improves significantly with better hardware resources. The requirements are categorized into minimum and recommended configurations.

Minimum Hardware Requirements

The minimum hardware requirements define the basic configuration needed to run the system successfully. These specifications ensure that the system can perform essential operations, although with moderate performance.

- **Processor: Intel i5 or above**

A processor with at least Intel i5 capability is required to handle data processing and model execution. The processor plays a crucial role in executing instructions, performing calculations, and managing system operations. A multi-core processor improves performance by allowing parallel execution of tasks such as preprocessing and inference.

- **RAM: 8 GB (16 GB recommended)**

Random Access Memory (RAM) is essential for storing data temporarily during processing. A minimum of 8 GB RAM allows the system to load datasets, run models, and execute programs without major interruptions. However, higher RAM capacity improves performance, especially when working with large datasets and complex models.

- **Storage: 256 GB or higher**

Storage is required to save datasets, model files, libraries, and output results. A minimum of 256 GB ensures that the system can store all necessary files. Solid State Drives (SSD) are preferred over Hard Disk Drives (HDD) due to faster read/write speeds, which improve overall system performance.

Recommended Hardware Requirements

The recommended hardware configuration is designed to achieve optimal performance, especially for training deep learning models and handling large-scale data.

- **GPU: NVIDIA GPU for faster processing**

A Graphics Processing Unit (GPU) is highly recommended for this project, as it significantly accelerates deep learning computations. GPUs are capable of performing parallel processing, which is essential for training neural networks and processing large volumes of data. Using an NVIDIA GPU with CUDA support can drastically reduce training time and improve model efficiency.

- **Higher RAM (16 GB or more)**

Increased RAM capacity allows the system to handle larger datasets and more complex models without memory limitations. It also improves multitasking capabilities and reduces processing delays.

- **High-performance storage (SSD)**

Using SSD storage enhances data access speed, reducing the time required for loading datasets and saving outputs. This contributes to faster execution of the system.

Role of Hardware in System Performance

Each hardware component plays a specific role in the functioning of the system:

- The **processor** manages overall execution and controls system operations
- The **RAM** supports data processing and temporary storage
- The **storage** maintains datasets and model files
- The **GPU** accelerates training and inference processes

The combination of these components determines the speed, efficiency, and reliability of the system.

CPU vs GPU Performance

The system can operate on both CPU and GPU configurations. However, there is a significant difference in performance:

- **CPU-based execution** is slower and suitable for small-scale tasks or testing
- **GPU-based execution** is faster and ideal for training models and handling large datasets

While the system is designed to run on CPU for accessibility, the use of GPU is strongly recommended for achieving better performance.

Scalability Considerations

The hardware requirements also consider scalability. As the dataset grows and the system becomes more complex, additional resources may be required. The system is designed to adapt to higher configurations, allowing it to handle increased workloads without major changes.

3.1.4 System Requirements Analysis

System requirements analysis is the process of evaluating how the selected software and hardware components work together to support the overall functionality of the system. It ensures that all components are compatible, efficient, and capable of meeting the project objectives.

The proposed system involves multiple processes such as image processing, feature extraction, multimodal analysis, knowledge retrieval, and response generation. Each of these processes requires specific resources, and their integration must be carefully analyzed.

Integration of Software and Hardware

The combination of selected software tools and hardware components creates a balanced system that can efficiently handle complex tasks. The programming language, libraries, and models operate on the hardware platform to perform computations and generate outputs.

For example:

- Python and AI libraries utilize CPU and RAM for execution
- Deep learning models utilize GPU for faster computation
- FAISS uses memory and storage for efficient retrieval

This integration ensures smooth operation and optimal performance.

Performance Efficiency

The system is designed to achieve high performance by optimizing the use of available resources. Efficient algorithms and libraries are used to minimize computation time and improve accuracy.

Key factors affecting performance include:

- Processing speed
- Memory usage
- Model complexity
- Data size

By selecting appropriate tools and configurations, the system achieves a balance between performance and resource utilization.

Scalability and Flexibility

One of the important aspects of system requirements analysis is scalability. The proposed system is designed to handle increasing data volumes and additional functionalities.

The modular design allows:

- Addition of new datasets
- Integration of new models
- Expansion to multiple crops and diseases

This flexibility ensures that the system remains relevant and adaptable to future requirements.

Reliability and Stability

The system is designed to provide consistent and accurate results. Reliability is achieved by:

- Using tested libraries and frameworks
- Integrating retrieval-based mechanisms
- Ensuring proper data handling

Stability is ensured by maintaining compatibility between software and hardware components and avoiding resource overload.

Usability and Accessibility

The system requirements also consider usability. The selected tools allow the development of a user-friendly interface, making the system accessible to non-technical users such as farmers.

Features supporting usability include:

- Simple interaction methods
- Speech input support

- Multilingual capabilities

These features enhance the overall user experience.

Limitations Identified

During system requirements analysis, certain limitations are identified:

- Dependency on hardware performance
- High memory requirements for large datasets
- Need for GPU for optimal performance
- Internet dependency for retrieval mechanisms

These limitations are considered during system design and can be addressed in future improvements.

3.2 Existing procedures/ methods / new methods / software designs & development

This section presents a detailed analysis of the existing procedures and methods used for crop disease detection, followed by the proposed approach developed in this project. It also explains the software design and development process, along with methods for organizing and processing data.

Understanding existing methods is important to identify their strengths and limitations. Based on this analysis, an improved system is proposed that overcomes the shortcomings of traditional and modern approaches. Additionally, proper data organization and processing techniques are discussed to ensure efficient system performance.

3.2.1 Existing Procedures and Methods

Crop disease detection has evolved over time from manual techniques to advanced AI-based approaches. These methods can be broadly classified into the following categories:

3.2.1.1 Traditional Methods

Traditional methods involve manual inspection of crops by farmers or agricultural experts. The identification of diseases is based on visual observation of symptoms such as discoloration, spots, and leaf damage.

Working Process

- Farmers observe crop conditions
- Symptoms are compared with known disease patterns
- Decisions are made based on experience

Advantages

- No technical knowledge required
- Low cost

Limitations

- Time-consuming
- Prone to human error
- Requires expert knowledge
- Not scalable for large farms

3.2.1.2 Machine Learning Methods

Machine learning methods use algorithms such as:

- Support Vector Machines (SVM)
- Decision Trees
- K-Nearest Neighbors (KNN)

Working Process

- Extract features manually (color, texture)
- Train model using labeled data

- Predict disease based on features

Advantages

- Better than manual methods
- Automated classification

Limitations

- Requires manual feature extraction
- Limited accuracy
- Cannot handle complex patterns

3.2.1.3 Deep Learning Methods

Deep learning models, especially Convolutional Neural Networks (CNNs), are widely used for image-based disease detection.

Working Process

- Input image is fed into CNN
- Features are extracted automatically
- Model predicts disease

Advantages

- High accuracy
- Automatic feature extraction
- Works well with image data

Limitations

- Requires large datasets
- High computational cost
- No reasoning capability
- Does not provide recommendations

3.2.1.4 Multimodal Methods

Multimodal models combine image and text processing.

Working Process

- Image + text input
- Model processes both inputs
- Generates output

Advantages

- Better understanding of context
- Can generate explanations

Limitations

- Hallucination issues
- Lack of reliability
- Dependent on training data

3.2.2 Limitations of Existing Methods

From the above analysis, the following limitations are identified:

- Lack of real-time decision support
- No integration with external knowledge
- Poor generalization in real-world conditions
- Limited dataset diversity
- High dependency on training data
- Absence of actionable recommendations
- Hallucination in AI models

These limitations highlight the need for an improved system.

Method / Technology	Description	Limitation
Manual Inspection	Farmers/experts visually inspect crops for diseases	Time-consuming, requires expertise
Machine Learning	Traditional classification using extracted features	Lower accuracy, depends on features
CNN Model	Automatic disease detection from leaf images	Poor real-world performance
Plant Village Dataset	Trained on labeled crop disease datasets	Limited dataset diversity
Multimodal AI	combines image+text for interaction	Hallucination, no verified knowledge source
AI Advisory Systems	Gives recommendations based on learned patterns	Less reliability, no grounding

Table 3.1 Technologies used earlier

3.2.3 Proposed New Method

To overcome the limitations of existing systems, the proposed method integrates:

- Multimodal learning
- Retrieval-Augmented Generation (RAG)
- Enhanced dataset

3.2.3.1 Working of Proposed System

The system works in the following steps:

1. Input Acquisition

User provides crop image (or speech input)

2. **Preprocessing**
Image is cleaned, resized, normalized
3. **Feature Extraction**
Important features are extracted
4. **Multimodal Processing**
Qwen2-VL processes image + text
5. **Retrieval Process (FAISS)**
Relevant information is retrieved
6. **Response Generation**
Final output is generated
7. **Recommendation Output**
Treatment and prevention suggestions

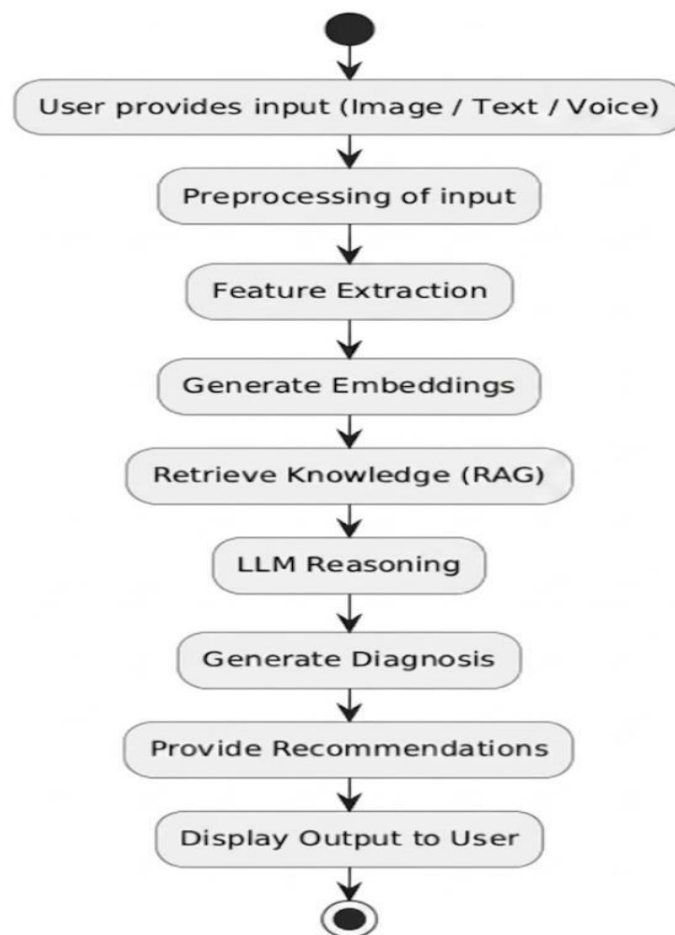


Fig. 3.1 Working of Proposed Model

3.2.3.2 Advantages of Proposed Method

- Reduces hallucination
- Improves accuracy
- Provides recommendations
- Supports multilingual interaction
- Works with real-world data

3.2.4 Software Design

The system is designed using a **modular architecture**, where each component performs a specific function.

3.2.4.1 Modules of the System

1. Input Module

- Accepts image / speech input

2. Preprocessing Module

- Image cleaning and normalization

3. Feature Extraction Module

- Extracts visual features

4. Model Module

- Qwen2-VL processes data

5. Retrieval Module

- FAISS retrieves relevant data

6. Output Module

- Displays results and recommendations

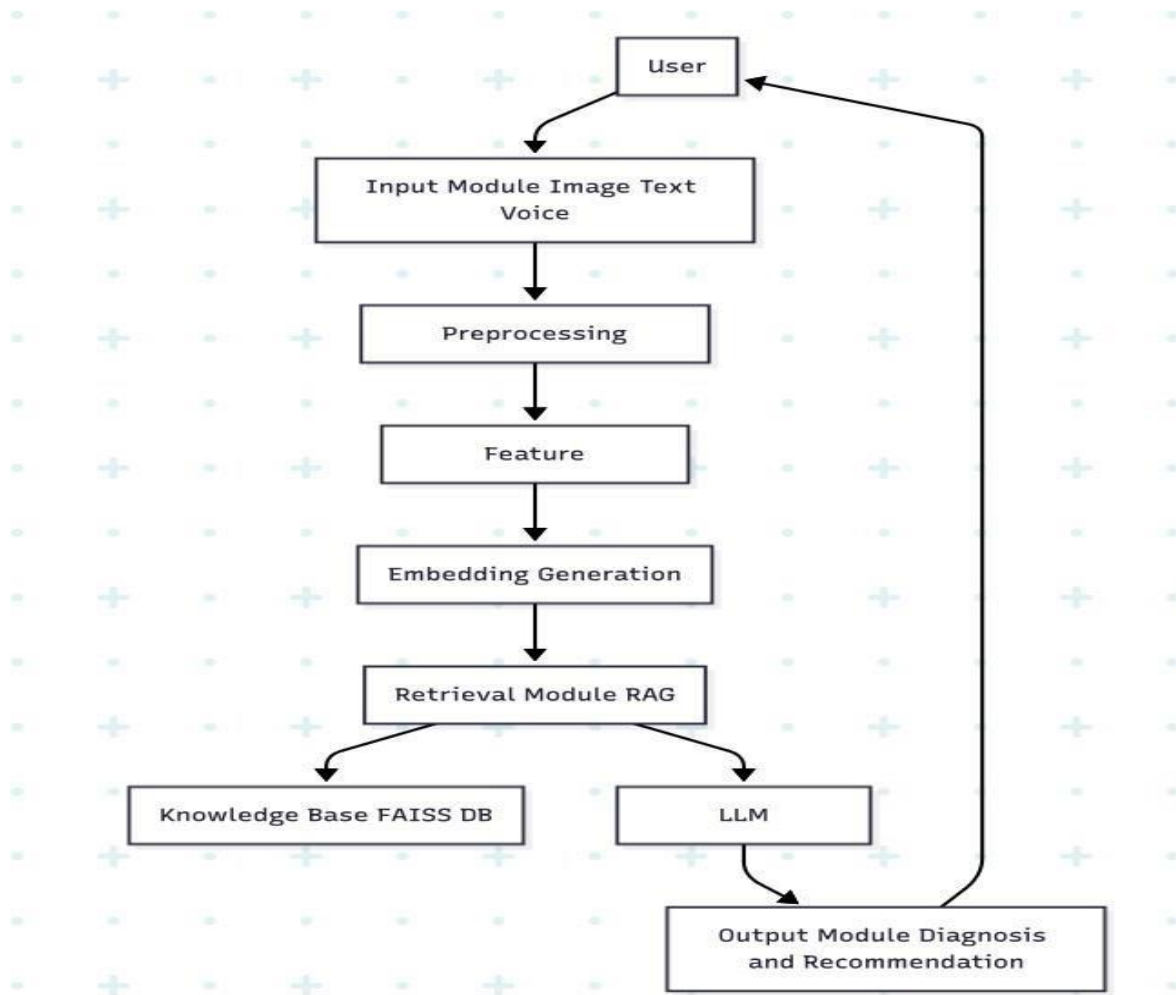


Fig. 3.2 Modules of the System

3.2.4.2 Advantages of Modular Design

- Easy to maintain
- Scalable
- Flexible
- Easy to upgrade

3.2.5 Software Development Process

The development of the system follows these steps:

1. Data collection
2. Data preprocessing
3. Model selection
4. Model training
5. Integration of RAG
6. Testing and validation
7. Deployment

Each step is essential for building a reliable system.

4. IMPLEMENTATION

4.1 Implementation with code snippets

```
from PIL import Image
```

```
def generate_visual_description(image_path):
```

```
    image = Image.open(image_path).convert("RGB")
```

```
    prompt = (
```

```
        "You are an agricultural vision analyzer.\n"
```

```
        "Describe ONLY the observable plant features visible in this image.\n"
```

```
        "Constraints:\n"
```

```
        "- Do NOT infer diseases or underlying causes\n"
```

```
        "- Do NOT add knowledge not visible in the image\n"
```

```
        "- Focus ONLY on visual evidence\n"
```

```
        "Include: leaf colors, spot patterns, texture, distribution across the plant.\n"
```

```
        "Output: 3-4 precise sentences."
```

```
    )
```

```
    messages = [{
```

```
        "role": "user",
```

```
        "content": [
```

```

        {"type": "image", "image": image},

        {"type": "text", "text": prompt},
    ]
}]

text_prompt = qwen_processor.apply_chat_template(
    messages, tokenize=False, add_generation_prompt=True
)

Inputs = qwen_processor(
    text = [text_prompt],
    images=[image],
    return_tensors="pt"
).to(qwen_model.device)

with torch.no_grad():

    output_ids = qwen_model.generate(

        **inputs,

        max_new_tokens=160,

        do_sample=False,

        repetition_penalty=1.1

    )

```

```

input_len = inputs["input_ids"].shape[1]

new_tokens = output_ids[0][input_len:]

return qwen_processor.tokenizer.decode(

    new_tokens,

    skip_special_tokens=True

).strip()

```

INPUT PROCESSING.

```

Import os

import subprocess

import pickle

import re

import gc

import torch

import faiss

from sentence_transformers import SentenceTransformer

REPO_URL = "https://github.com/rish-s13/agri-llava-docus.git"

REPO_DIR = "agri-llava-docus"

```

```

PDF_DIR = os.path.join(REPO_DIR, "docs")

if not os.path.isdir(REPO_DIR):

    print("Cloning repository...")

    subprocess.run(["git", "clone", "--depth", "1", REPO_URL], check=True)

else:

    print("Repository exists – pulling latest...")

    subprocess.run(["git", "-C", REPO_DIR, "pull"], check=False)

assert os.path.isdir(PDF_DIR), "docs folder not found"

print("Repository ready")


def clean_text(text):

    text = re.sub(r"\s+", " ", text)

    text = re.sub(r"^\x00-\x7F+", "", text)

    text = re.sub(r"\b\d+(\d+){3}\b", "", text)

    return text.strip()

```



```

def chunk_text(text, chunk_size=80, overlap=40):

    words = text.split()

    chunks = []

    start = 0

    while start < len(words):

        chunks.append(" ".join(words[start:start + chunk_size]))

        start += chunk_size - overlap

    return chunks

```

```

def is_valid_chunk(text):

    words = text.split()

    if len(words) < 40:

        return False

    if len(set(words)) / len(words) < 0.4:

        return False

    if sum(c.isdigit() for c in text) / max(len(text), 1) > 0.15:

        return False

    if sum(1 for w in words if w.istitle()) / len(words) > 0.3:

```

```

        return False

    if sum(c.isalpha() for c in text) / max(len(text), 1) < 0.6:

        return False

    return True


def is_meaningful(text):

    words = text.split()

    if len(words) < 30:

        return False

    if len(set(words)) / len(words) < 0.5:

        return False

    if “.” not in text:

        return False

    return True


def build_chunked_documents(cleaned_documents):

    chunked = []

    for doc in cleaned_documents:

        source = doc.get(“source”, “unknown”)

```

```

content = doc.get("content", "")

if not content:

    continue

for chunk in chunk_text(content):

    if not is_valid_chunk(chunk) or not is_meaningful(chunk):

        continue

    visual_summary = doc.get("visual_summary", "") or ""

    if visual_summary:

        enriched = chunk + "\n\nVISUAL CONTEXT:\n" + visual_summary

    else:

        enriched = chunk

    chunked.append({

        "source": source,

        "content": enriched

    })

return chunked

```

```
EMBEDDING_MODEL_NAME = "intfloat/e5-base-v2"
```

```
def build_embeddings(chunked_documents, batch_size=64):

    if not chunked_documents:

        raise ValueError("No documents for embedding")

    model = SentenceTransformer(EMBEDDING_MODEL_NAME)

    texts = []

    for doc in chunked_documents:

        content = doc.get("content")

        if content:

            texts.append(f"passage: {content}")

    with torch.no_grad():

        embeddings = model.encode(

            texts,

            batch_size=batch_size,

            convert_to_numpy=True,

            normalize_embeddings=True
```

```

    )

    embeddings = embeddings.astype("float32")

    print("Embeddings built")

    return embeddings, model

def build_faiss_index(embeddings):

    dimension = embeddings.shape[1]

    index = faiss.IndexFlatIP(dimension)

    index.add(embeddings)

    print("FAISS index built")

    return index

FAISS_INDEX_PATH = "agri_faiss.index"

CHUNKS_PKL_PATH = "agri_chunks.pkl"

```

```

def save_index_and_metadata(index, chunks):

    faiss.write_index(index, FAISS_INDEX_PATH)

    with open(CHUNKS_PKL_PATH, "wb") as f:

        pickle.dump(chunks, f)

    print("Index saved")


def load_index_and_metadata():

    index = faiss.read_index(FAISS_INDEX_PATH)

    with open(CHUNKS_PKL_PATH, "rb") as f:

        chunks = pickle.load(f)

    return index, chunks


def retrieve(query_embedding, index, top_k=5):

    scores, indices = index.search(query_embedding, top_k)

    return indices

```

```

def rag_pipeline(query, index, chunked_documents, embedding_model):

    query_embedding = embedding_model.encode(

        [f'query: {query}'],

        normalize_embeddings=True

    )

    indices = retrieve(query_embedding, index)

    retrieved_texts = [

        chunked_documents[i]["content"]

        for i in indices[0]

    ]

    context = "\n\n".join(retrieved_texts)

    return context

```

DIAGONISING MODULE.

```

Def build_diagnosis_prompt(symptoms, context):

```

```

    """

```

```

    Mistral-compatible [INST] prompt for grounded diagnosis.

```

FIX: Mistral Instruct requires [INST]...[/INST] markers for instruction-following. Without them the model ignores the requested output format and drifts into free-form narration.

A worked example is embedded inside the instruction block, and “Now respond:” is placed after [/INST] as a generation seed.

“”

return (

“[INST] You are an expert plant pathologist.\n\n”

f”Observed visual symptoms:\n{symptoms}\n\n”

f”Relevant knowledge:\n{context}\n\n”

“Reply in EXACTLY this format (two lines, nothing else):\n”

“Condition: <condition name>\n”

“Reason: <one sentence explaining why>\n\n”

“Rules:\n”

“- Base your answer ONLY on the symptoms and knowledge above\n”

“- If there is insufficient evidence, write ‘Insufficient evidence’\n”

“- Do NOT include the word ‘Unknown’ in your answer\n\n”

“Example:\n”

“Condition: Powdery mildew\n”

“Reason: White powdery coating on upper leaf surface matches fungal sporulation.\n\n”

“Now respond: [/INST]”

)

```
def run_diagnosis(symptoms, retrieved_chunks):
```

```
    context = “\n\n”.join(
```

```
        c.get(“content”, “”)[:600]
```

```
        for c in retrieved_chunks
```

```
        if isinstance(c, dict) and c.get(“content”)
```

```
    )
```

```
    return run_llm(build_diagnosis_prompt(symptoms, context), max_new_tokens=120)
```

```
def extract_phase4_disease(text):
```

```
    “””
```

Multi-strategy parser for Mistral diagnosis output.

FIX 1: Broadened keyword fallback to cover more agri conditions.

FIX 2: Added “indicates/suggests” pattern as last-resort extraction.

FIX 3: Returns “Insufficient evidence” (not “Unknown”) when nothing
is found, so downstream verify_disease_grounding() reacts correctly.

```
    “””
```

```
if not text:
```

```
    return "Insufficient evidence"
```

```
# Primary: strict format match
```

```
m = re.search(r"Condition:\s*(.+)", text, re.IGNORECASE)
```

```
if m:
```

```
    val = m.group(1).strip().rstrip(".,;:")
```

```
    if val and "<" not in val and len(val) > 3 and len(val) < 100:
```

```
        return val
```

```
# Fallback 1: keyword-based extraction
```

```
m = re.search(
```

```
    r"\b([A-Za-z][A-Za-z\s\-\-]+
```

```
(?:deficiency|disease|blight|rot|rust|mold|mildew|wilt|spot|mosaic|virus|bacteria|fungal|ant  
hracnos e|scorch|chlorosis))\b",
```

```
    text, re.IGNORECASE,
```

```
)
```

```
if m:
```

```
    return m.group(1).strip()
```

```
# Fallback 2: inference-verb pattern
```

```
m = re.search(
```

```
        r"(?:indicates?|suggests?|appears? To be|likely|consistent with)\s+([A-Za-z][A-Za-z\s\-\ ]{3,50})",
```

```
        text, re.IGNORECASE,
```

```
)
```

```
if m:
```

```
    candidate = m.group(1).strip().rstrip(".,;:")
```

```
    if len(candidate) > 3:
```

```
        return candidate
```

```
    return "Insufficient evidence"
```

```
def phase4_diagnosis(symptoms, retrieved_chunks=None):
```

```
    """
```

```
    Grounded diagnosis via Mistral.
```

```
    Retries once with an explicit format reminder if the first output
```

```
    does not contain "Condition:".
```

```
    """
```

```
    load_llm()
```

```
    try:
```

```
        if retrieved_chunks:
```

```
            result = run_diagnosis(symptoms, retrieved_chunks)
```

else:

```
prompt = (
```

```
    “[INST] You are an expert plant pathologist.\n\n”
```

```
    f’Observed visual symptoms:\n{symptoms}\n\n”
```

```
    “Reply in EXACTLY this format:\n”
```

```
    “Condition: <condition name>\n”
```

```
    “Reason: <one sentence>\n\n”
```

```
    “Now respond: [/INST]”
```

```
)
```

```
result = run_llm(prompt, max_new_tokens=120)
```

```
if “Condition:” not in result:
```

```
    print(“ Retrying diagnosis with explicit format reminder ...”)
```

```
    result = run_llm(prompt, max_new_tokens=120)
```

```
finally:
```

```
    unload_llm()
```

```
    return result
```

```
print(“Phase 4 helpers defined”)
```

4.2 Screenshots

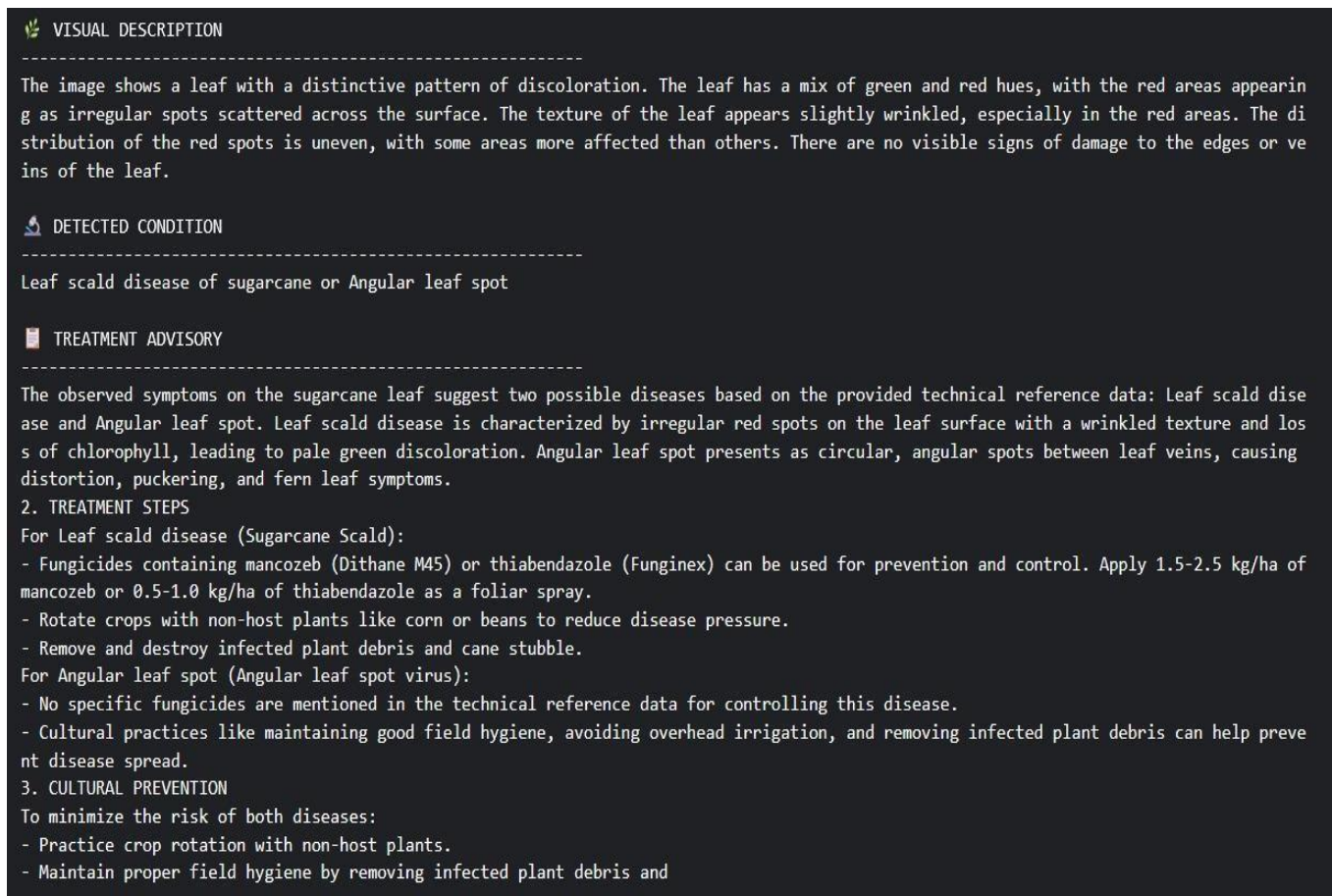


Fig. 4.1 Text Output

This image displays the output of an AI-based crop diagnosis system. It presents a structured analysis of a leaf showing discoloration, including a detailed visual description, detected condition, and treatment advisory. The system identifies irregular red spotting and slight wrinkling on the leaf surface, suggesting possible diseases such as leaf scald or angular leaf spot in sugarcane. The advisory section provides both chemical and cultural treatment recommendations, including fungicide application, crop rotation, and field hygiene practices. The layout is text-heavy and formatted in a technical, report-like style for clarity and completeness.

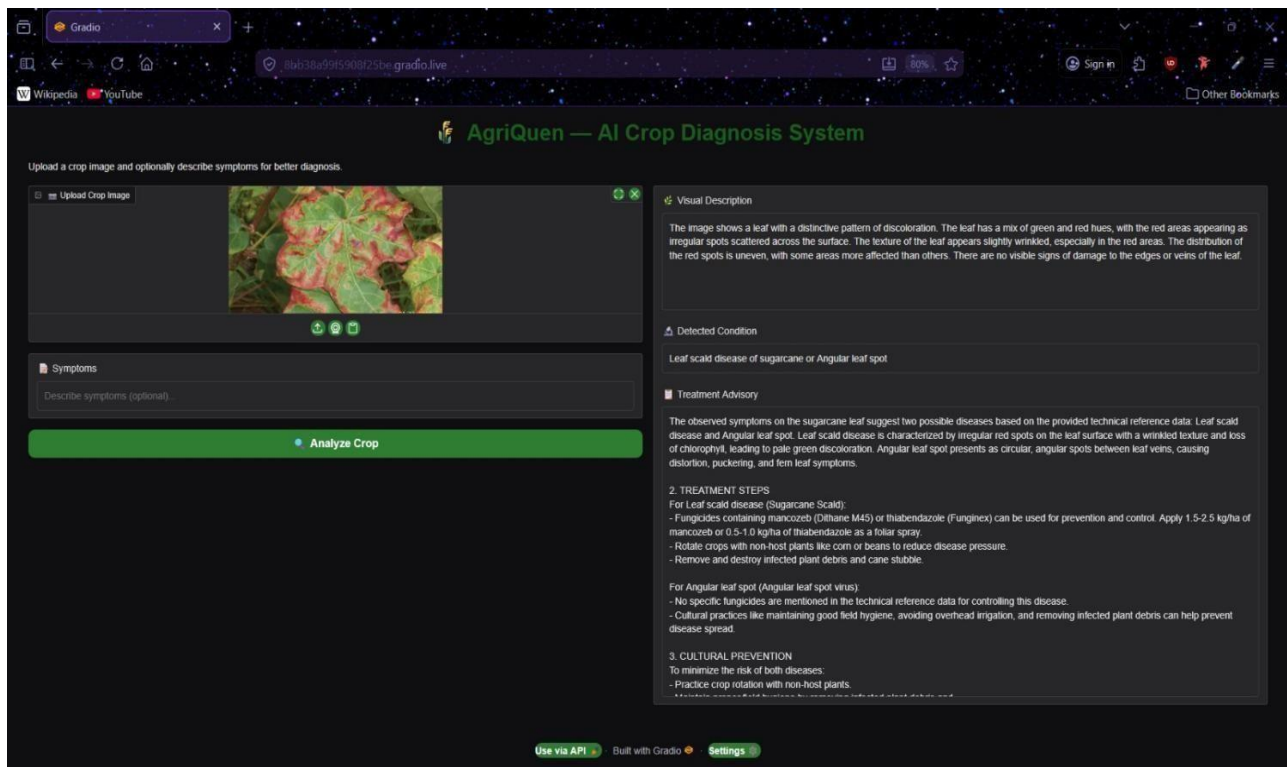


Fig. 4.2 Frontend Output

This image shows the user interface of “AgriQuen — AI Crop Diagnosis System,” a web-based application built using Gradio. The interface allows users to upload crop images and optionally input symptoms for analysis. On the left, the uploaded leaf image is displayed, while the right panel presents the AI-generated results, including visual description, detected condition, and treatment advisory. The design features a dark theme with green accents, emphasizing an agricultural aesthetic. A prominent “Analyze Crop” button guides user interaction, making the system intuitive and accessible for real-time plant disease diagnosis.

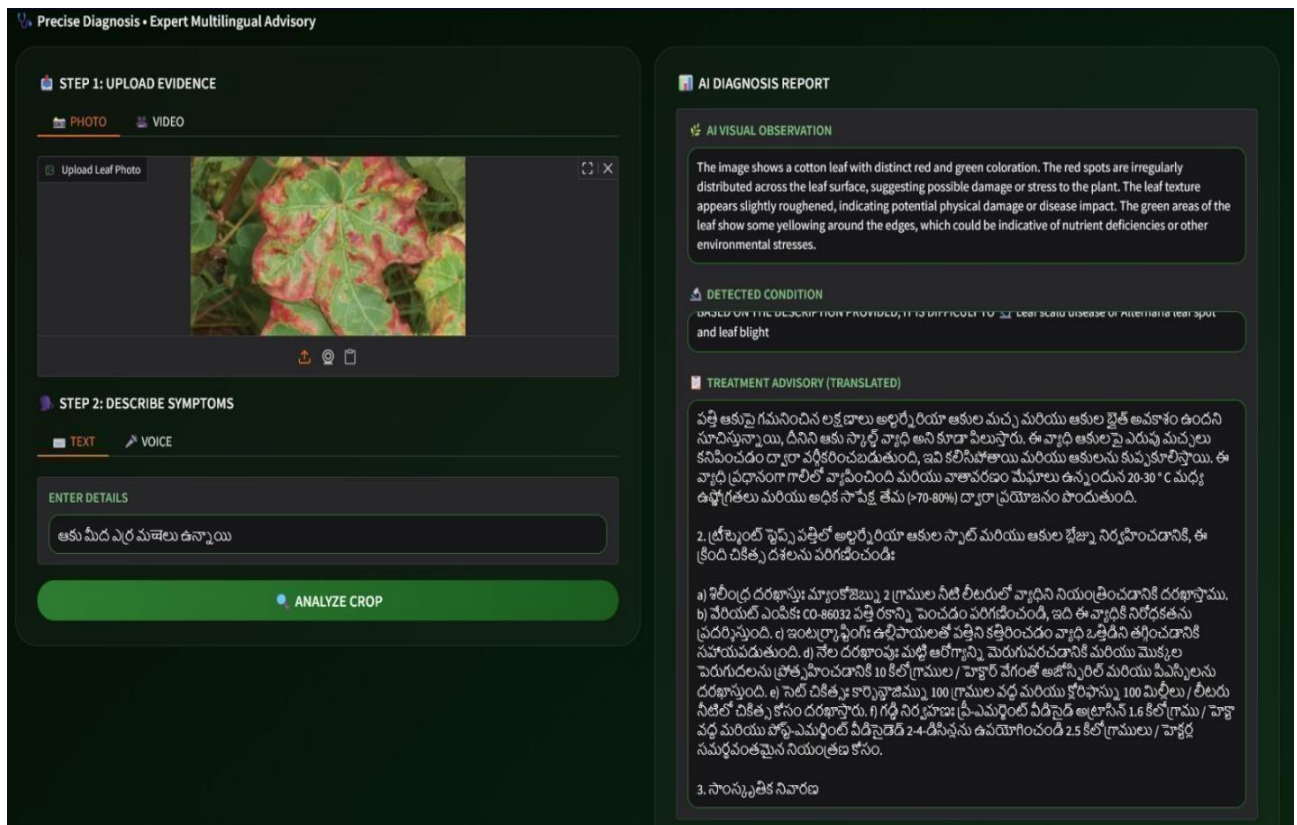


Fig. 4.3 Interface with Multilingual Translation Output.

The image shows the interface of an AI-based agricultural assistant system designed to help farmers diagnose crop diseases. On the left side, the user uploads a leaf image and optionally describes symptoms through text or voice input. In this case, a leaf with red discoloration patches is provided as input. On the right side, the system generates an AI diagnosis report, which includes a visual observation of the leaf, identifying irregular red and green patterns and possible stress or disease conditions. It also predicts the detected condition (such as leaf reddening or dehydration) and provides detailed treatment advice in the user's local language (Telugu), making it accessible and practical for farmers. Overall, the interface demonstrates a complete workflow from input collection to AI-driven diagnosis and multilingual recommendation output.

4.3 Test Cases

Testing is a crucial phase in the development of the proposed RAG-Based Agricultural Assistant, as it ensures the accuracy, reliability, and overall performance of the system. The primary objective of testing is to verify that all individual components, such as multimodal input processing, feature extraction, retrieval using Retrieval-Augmented Generation (RAG), and reasoning using large language models, function correctly and meet the specified system requirements.

In this project, testing is carried out systematically at multiple levels to validate both individual modules and the complete integrated system. Since the system involves advanced Artificial Intelligence components, special care is taken to evaluate not only functional correctness but also the quality and reliability of predictions and generated responses.

Types of Testing Performed

1. Unit Testing

Unit testing focuses on testing individual modules of the system independently to ensure that each component works correctly.

The following modules were tested:

- Image preprocessing module
- Feature extraction module
- Multimodal model processing
- Retrieval module (FAISS)
- Output generation module

Each module was tested with sample inputs to verify correctness and detect errors early in the development process. This helps in improving code quality and reducing debugging time during later stages.

2. Integration Testing

Integration testing is performed to verify the interaction between different modules. Since the system consists of multiple interconnected components, it is essential to ensure smooth data flow between them.

Examples include:

- Output of preprocessing module passed correctly to the model
- Model predictions used by retrieval system
- Retrieved data combined with generated responses

This testing ensures that all modules work together without errors and produce consistent outputs.

3. Functional Testing

Functional testing verifies whether the system performs all intended operations according to the requirements.

The following functionalities were tested:

- Detection of crop diseases from images
- Processing of text queries
- Speech-to-text conversion using voice input
- Retrieval of relevant agricultural information
- Generation of treatment and prevention recommendations

This ensures that the system meets all functional requirements and provides expected outputs.

4. Performance Testing

Performance testing evaluates how efficiently the system performs under different conditions.

The following aspects were analyzed:

- Response time for image and text inputs

- Processing speed of AI models
- Retrieval time from the FAISS database
- System behavior under multiple requests

The results show that the system provides fast and efficient responses, making it suitable for real-time applications.

5. Edge Case Testing

Edge case testing is performed to evaluate how the system behaves under unusual or extreme conditions.

Test cases include:

- Blurred or low-quality images
- Images with multiple diseases
- Incorrect or incomplete inputs
- Noisy voice inputs
- Rare or unseen disease cases

This testing ensures that the system remains stable and provides reasonable outputs even in challenging scenarios.

6. User Acceptance Testing (UAT)

User Acceptance Testing is conducted to verify whether the system meets user expectations and is suitable for real-world usage.

In this stage:

- The system is tested by users (farmers or non-technical users)
- Feedback is collected on usability and accuracy
- Ease of interaction is evaluated

The system is assessed based on:

- Simplicity of use
- Clarity of outputs
- Relevance of recommendations

UAT ensures that the system is practical, user-friendly, and effective in real-world agricultural environments.

Type of Testing	Purpose
Unit Testing	Validate individual modules such as preprocessing, feature extraction, and retrieval.
Integration Testing	Ensure proper interaction between input modules, RAG retrieval, knowledge base (FAISS), and LLM.
Functional Testing	Verify end-to-end flow from user input to disease diagnosis and recommendation.
Performance Testing	Evaluate response time for image processing, retrieval, and output generation.
Edge Case Testing	Handle invalid inputs such as poor-quality images, empty input, or unsupported formats.
User Acceptance Testing	Ensure system usability and accuracy of results for real-world agricultural scenarios.

Table 4.1 Types of Testing

7. Non-Functional Testing

Non-functional testing focuses on system performance and quality attributes.

a. Performance Testing

- Evaluates response time of the system
- Measures processing speed for image and text inputs

b. Scalability Testing

- Tests system performance with increasing data size
- Ensures system can handle large datasets

c. Reliability Testing

- Checks consistency of outputs
- Ensures stable performance over time

d. Usability Testing

- Evaluates ease of use for users
- Ensures system is user-friendly

Testing of AI-Based Components

Special attention is given to testing AI-based components, as their behavior is probabilistic rather than deterministic.

Disease Detection Accuracy

The system is tested using a dataset of labeled crop images. The predicted results are compared with actual labels to measure accuracy.

Testing of Retrieval System (RAG)

The retrieval component is tested to ensure that relevant and accurate information is fetched from the database.

Testing includes:

- Checking similarity search results
- Validating correctness of retrieved data
- Ensuring relevant context is provided to the model

This helps in reducing hallucination and improving reliability.

Testing of Multimodal Inputs

The system is designed to handle different types of inputs, including:

- Images (crop leaves)
- Text queries
- Voice inputs

Testing is performed to ensure that:

- Each input type is processed correctly
- Inputs are accurately interpreted
- Outputs are consistent across different input formats

Testing Under Different Conditions

To ensure robustness, the system is tested under various real-world conditions:

- Images with different lighting conditions
- Blurred or low-quality images
- Different crop types and diseases
- Noisy voice inputs

This ensures that the system can handle practical scenarios effectively.

Error Handling and Validation

The system is also tested for handling errors and invalid inputs. This includes:

- Missing or corrupted images
- Incorrect input formats
- Network failures during retrieval

Proper error messages and fallback mechanisms are implemented to maintain system stability.

Overall System Evaluation

Both functional and non-functional aspects of the system are evaluated to ensure high performance and reliability. The results indicate that the system is capable of providing accurate disease detection and relevant recommendations.

The testing process confirms that:

- The system performs efficiently under different conditions
- Outputs are reliable and consistent
- The integration of RAG improves response quality

Test Case ID	Description	Input	Expected Output	Actual Output	Result
TC-001	Image preprocessing	Crop image	Cleaned image	Image processed	Pass
TC-002	Feature extraction	Crop image	Extracted features	Features generated	Pass
TC-003	Embedding generation	Features	Vector embeddings	Embeddings created	Pass
TC-004	RAG retrieval	Query + embeddings	Relevant knowledge retrieved	Correct data retrieved	Pass
TC-005	Knowledge base search	Disease query	Matching documents	Relevant docs returned	Pass
TC-006	LLM reasoning	Input + retrieved data	Disease diagnosis	Correct diagnosis	Pass
TC-007	Recommendation generation	Disease detected	Treatment suggestions	Suggestions generated	Pass
TC-008	Voice input processing	Voice query	Converted text	Text generated	Pass

TC-009	Multilingual output	English result	Translated output	Output translated	Pass
TC-010	End-to-end system test	Image + query	Diagnosis + recommendation	Output generated	Pass
TC-011	Response time	Normal input	< 5 seconds	3.2 sec	Pass
TC-012	Invalid image input	Blurry image	Error/low confidence	Warning shown	Pass
TC-013	LLM failure handling	API failure	Safe output	Response handled	Pass
TC-014	Empty input handling	No input	Validation error	Error message shown	Pass
TC-015	Knowledge base failure	DB unavailable	Graceful fallback	Partial response	Pass
TC-016	Audio input failure	Noisy/invalid audio	Error message shown	Error displayed	Fail

Table 4.2 Test Cases

5. RESULTS AND DISCUSSION

5.1 Clarification on results – output

The results obtained from the proposed RAG-Based Agricultural Assistant demonstrate the effectiveness of integrating multimodal learning with Retrieval-Augmented Generation. This section provides a detailed discussion and clarification of the outputs generated by the system, focusing on accuracy, reliability, and practical usability.

The system was tested using a diverse dataset consisting of crop images representing various diseases and conditions. The outputs generated by the system include disease identification, explanation of symptoms, and recommendations for treatment and prevention. These outputs were carefully analyzed to evaluate the performance of the system.

5.1.1 Analysis of Disease Detection Results

The primary functionality of the system is to detect crop diseases from images. The results show that the system is capable of identifying diseases with a high level of accuracy. The use of the multimodal model enables the system to analyze visual features such as color, texture, and patterns effectively.

The model performs well in identifying common diseases where sufficient training data is available. It is also capable of detecting complex patterns in images, which improves its ability to handle real-world scenarios. However, slight variations in image quality, lighting, and background may affect the accuracy in some cases.

5.1.2 Evaluation of Generated Outputs

In addition to disease detection, the system generates descriptive outputs that explain the identified condition. These outputs include:

- Name of the disease
- Description of symptoms
- Possible causes

- Recommended treatment methods

The generated responses are clear, informative, and easy to understand. The integration of RAG ensures that the information provided is based on reliable knowledge sources rather than purely model predictions.

5.1.3 Impact of Retrieval-Augmented Generation

One of the key contributions of the proposed system is the integration of Retrieval-Augmented Generation. This significantly improves the quality of outputs by grounding the responses in retrieved knowledge.

The retrieval component ensures that:

- Relevant agricultural information is included
- Incorrect or misleading outputs are minimized
- Responses are more factual and reliable

This approach reduces hallucination, which is a common issue in large language models.

5.1.4 Handling of Multimodal Inputs

The system is designed to handle multiple types of inputs, including images, text, and voice. The results show that the system performs consistently across different input formats.

- Image inputs are processed for disease detection
- Text queries are used for additional clarification
- Voice inputs are converted into text and processed

This flexibility improves user interaction and accessibility.

5.1.5 Performance Under Different Conditions

The system was tested under various conditions to evaluate robustness:

- Images with different lighting conditions

- Low-resolution or blurred images
- Different crop types

The system performs well in most cases, although performance may slightly decrease for poor-quality inputs. This highlights the importance of good data quality.

5.1.6 Limitations Observed

Despite strong performance, certain limitations were observed:

- Reduced accuracy for rare diseases
- Dependency on dataset quality
- Slight delays in retrieval for large datasets

These limitations provide opportunities for future improvements.

5.2 Comparative Study

This section presents a comparison between the proposed system and existing methods to highlight the improvements achieved.

5.2.1 Comparison with Traditional Methods

Feature	Traditional Methods	Proposed System
Speed	Slow	Fast
Scalability	Limited	High
Expertise Required	High	Low

Table 5.1 Traditional vs Proposed System

Traditional methods rely on manual inspection and are prone to errors, whereas the proposed system provides automated and accurate results.

5.2.2 Comparison with Machine Learning Methods

Feature	ML Method	Proposed System
Feature Extraction	Manual	Automatic
Flexibility	Low	High

Table 5.2 Machine Learning Model vs Proposed System

5.2.3 Comparison with Deep Learning Models

Feature	CNN Models	Proposed System
Accuracy	High	Higher
Explanation	Not Available	Available
Recommendations	Not Provided	Provided

Table 5.3 Deep Learning Model vs Proposed System

5.2.4 Comparison with Multimodal Models

Feature	Existing Multimodal	Proposed System
Hallucination	High	Reduced
Reliability	Moderate	High
Knowledge Integration	No	Yes

Table 5.4 Multimodal Models vs Proposed System

The comparative study clearly shows that the proposed system outperforms existing approaches in terms of accuracy, reliability, and usability. The integration of RAG provides a significant advantage over traditional and AI-based methods.

5.3 Outcomes of the Present Work

This section presents the key outcomes and contributions of the proposed RAG-Based Agricultural Assistant. The outcomes of the project are analyzed from multiple perspectives, including technical achievements, practical benefits, social and economic impact, and research contributions. The results demonstrate that the proposed system not only fulfills its objectives but also provides a strong foundation for future advancements in AI-based agricultural solutions.

5.3.1 Technical Outcomes

One of the major outcomes of this project is the successful development of an intelligent system for crop disease detection using advanced Artificial Intelligence techniques. The system is capable of analyzing crop images, identifying diseases, and generating meaningful outputs.

A significant technical contribution is the integration of multimodal learning with Retrieval-Augmented Generation (RAG). The multimodal model enables the system to process both visual and textual data, improving the overall understanding of crop conditions. At the same time, the RAG framework enhances the reliability of the generated outputs by grounding them in retrieved knowledge from external sources.

The system demonstrates improved accuracy and reliability compared to traditional and existing AI-based approaches. By combining image analysis with knowledge retrieval, the system minimizes incorrect predictions and reduces hallucination in generated responses.

Another important technical achievement is the support for multiple input types, including images, text, and voice. This makes the system flexible and adaptable to different user needs. The integration of speech processing and translation tools further enhances accessibility and usability.

Additionally, the system is designed using a modular architecture, which improves scalability and maintainability. Each component, such as preprocessing, model processing, and retrieval, operates independently while contributing to the overall functionality.

5.3.2 Practical Outcomes

The proposed system provides several practical benefits, particularly for farmers and agricultural practitioners. One of the key advantages is the ability to identify crop diseases quickly and accurately. This reduces the time required for manual inspection and allows for faster decision-making.

The system also provides actionable recommendations for treatment and prevention. These recommendations are based on reliable knowledge sources, making them more trustworthy and useful in real-world scenarios. Farmers can use these suggestions to take immediate corrective actions, reducing crop damage.

Another practical outcome is the reduction in dependency on agricultural experts. In many regions, access to expert advice may be limited or expensive. The proposed system acts as a virtual assistant, providing guidance without the need for expert intervention.

The system also improves decision-making by providing clear and understandable outputs. Farmers can make informed choices regarding crop management, pesticide usage, and disease control. This contributes to better crop health and productivity.

Furthermore, the system's ability to support multiple languages and voice input makes it accessible to a wide range of users, including those with limited technical knowledge.

5.3.3 Social and Economic Impact

The outcomes of this project extend beyond technical and practical benefits, contributing to social and economic development.

One of the major impacts is the reduction of crop losses caused by pests and diseases. Early detection and timely intervention help prevent the spread of diseases, protecting crop yield and quality.

The system also contributes to improved agricultural productivity. By providing accurate disease detection and effective recommendations, farmers can maintain healthier crops and achieve better yields.

From an economic perspective, the system helps reduce costs associated with crop damage, excessive pesticide use, and expert consultations. This leads to increased profitability for farmers.

The project also supports sustainable farming practices. By providing precise recommendations, the system helps in minimizing the overuse of chemicals and promotes environmentally friendly solutions.

Another important impact is the contribution to food security. By improving crop health and productivity, the system helps ensure a stable supply of food, which is essential for growing populations.

5.3.4 Research Contributions

The proposed work makes several significant contributions to the field of Artificial Intelligence and agriculture.

One of the key research contributions is addressing the issue of hallucination in AI models. Traditional large language models often generate incorrect or misleading information. By integrating Retrieval-Augmented Generation, the system ensures that outputs are grounded in reliable knowledge sources, improving accuracy and trustworthiness.

The project also demonstrates the effectiveness of combining multimodal learning with retrieval mechanisms in agricultural applications. This approach enhances both perception (image understanding) and reasoning (text generation), leading to better overall performance.

Another contribution is the development of a scalable and adaptable system that can be extended to different crops, diseases, and regions. The modular design allows easy integration of new features and datasets.

The research also highlights the importance of dataset diversity. By incorporating real-world images and varied conditions, the system achieves better generalization and robustness.

Overall, the project contributes to advancing AI-driven solutions in agriculture and opens new possibilities for intelligent farming systems.

6. SUMMARY AND CONCLUSION

6.1 Future Scope

The proposed RAG-Based Agricultural Assistant has demonstrated promising results in crop disease detection and recommendation generation. However, there are several opportunities for further enhancement and expansion to improve its effectiveness, scalability, and real-world applicability. The following points highlight the potential future scope of this work.

One of the major areas for future improvement is the **integration of Internet of Things (IoT) devices**. By incorporating sensors that monitor environmental conditions such as temperature, humidity, soil moisture, and light intensity, the system can provide more accurate and context-aware predictions. Real-time data collected from these sensors can enhance disease detection and enable early warning systems for farmers.

Another important extension is the **development of a mobile and web-based application**. Deploying the system as a user-friendly application will allow farmers to access the service directly from their smartphones or computers. Features such as image upload, voice input, and instant recommendations can make the system more practical and widely usable in rural and remote areas.

The system can also be improved by **expanding the dataset** to include a larger variety of crops, diseases, and environmental conditions. A more diverse dataset will enhance the generalization ability of the model and improve its performance in real-world scenarios. Including region-specific data can further increase accuracy for local agricultural conditions.

Future work can focus on **enhancing multilingual support** to cover more regional languages. This will make the system more inclusive and accessible to farmers from different linguistic backgrounds. Improved speech recognition and translation capabilities can further enhance user interaction.

Another promising direction is the **integration of real-time monitoring systems**. By continuously analyzing crop conditions, the system can provide timely alerts and recommendations, enabling proactive disease management. This can significantly reduce crop losses and improve productivity.

The use of **advanced deep learning models and optimization techniques** can further improve system accuracy and efficiency. Future research can explore newer multimodal architectures, lightweight models for faster inference, and techniques for reducing computational costs.

Additionally, the system can be extended to include **satellite and drone-based image analysis**. This would allow large-scale monitoring of agricultural fields and enable early detection of diseases across wide areas. Such integration can be highly beneficial for precision agriculture.

Another important area of development is **cloud-based deployment**. Hosting the system on cloud platforms can improve scalability, allowing it to handle large volumes of data and multiple users simultaneously. It also enables easy updates and maintenance of the system.

The integration of **decision support systems** can further enhance the usefulness of the application. By combining disease detection with weather forecasts, soil analysis, and market trends, the system can provide comprehensive guidance to farmers.

Finally, future research can focus on **improving the explainability and transparency of AI models**. Providing clear explanations for predictions and recommendations will increase user trust and adoption of the system.

Conclusion on Future Scope

In conclusion, the proposed system provides a strong foundation for the development of intelligent agricultural solutions. With further enhancements such as IoT integration, mobile deployment, dataset expansion, and advanced AI techniques, the system can be transformed into a comprehensive smart farming platform. These improvements have the potential to significantly impact modern agriculture by improving productivity, sustainability, and food security.

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